1

Introduction to Astronomy

In this chapter, we will learn about

- the basic features and hierarchy of celestial bodies
- the basic units in astronomy
- the geocentric and heliocentric models
- Galileo's astronomical discoveries

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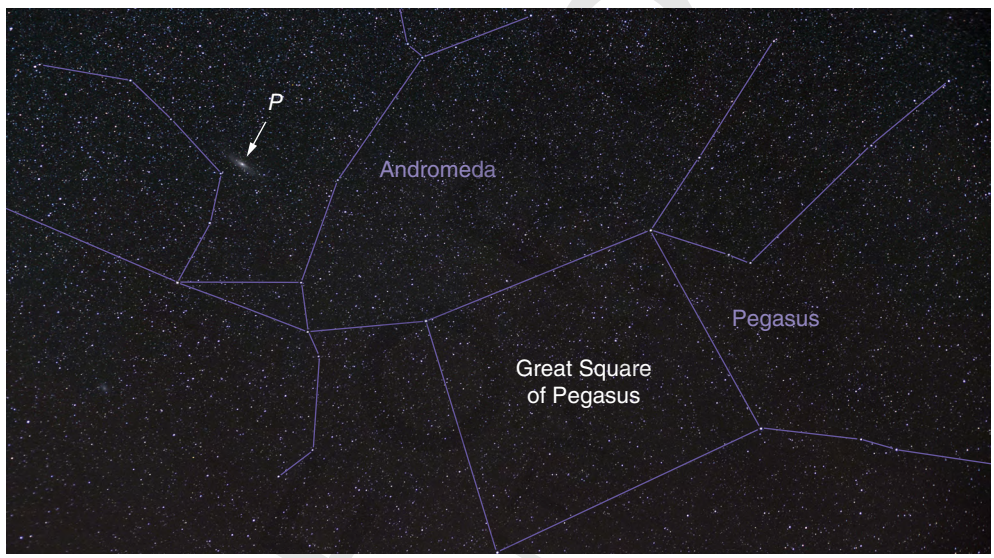
1.1 Layers of structure of the universe

People often call all the light dots in the night sky collectively as 'stars'. However, when you look at Figure 1.1a with very careful attention, you may notice that *P* is not just a light dot.



The *Great Square of Pegasus*, which consists of four stars of similar brightness, is a prominent sign in the autumn night sky of the northern hemisphere.

A group of stars appears to form a pattern when joined by imaginary lines. The region of sky containing the pattern is called a *constellation*. The whole sky has 88 constellations now.

Fig 1.1a *Constellations of Pegasus and Andromeda in the night sky.*

Spot *P* appears as a faint, hazy patch of light. In fact, it is not a single star but a collection of a trillion (10^{12}) of stars (Fig 1.1b)!

Fig 1.1b *The magnified view of spot P in Figure 1.1a.*

✎ Spot *P* is the Andromeda Galaxy (仙女座星系). It is the nearest spiral galaxy outside the Milky Way Galaxy and the largest galaxy in the Local Group. It is catalogued as M31 or NGC 224. Since the Andromeda Galaxy has an apparent magnitude of 3.4, it is easily visible to the naked eye.

DSE goal

Describe the basic features of celestial bodies such as satellite, planet, star, star cluster, nebula, galaxy and cluster of galaxies.

DSE exam

16(2)Q1.3, 20(2)Q1.1

1 Celestial bodies

The night sky is more than just the moon and stars. There are many more **celestial bodies** in it. We will go through some of them below.

a Stars, planets and satellites

Stars are celestial bodies that generate energy from nuclear fusion. They give out heat and light on their own. The Sun is a typical star and the closest star to us.

Great Square of Pegasus 秋季大四方　constellation 星座　celestial body 天體　star 恆星

The Earth where we live on is a **planet**. It orbits around the Sun with seven other planets. The Sun and the eight planets are the major parts of the *solar system* (Fig 1.1c).

There are also smaller planet-like celestial bodies, called *dwarf planets*, orbiting around the Sun. Pluto is one of them. It was once called the ninth planet but reclassified as dwarf planet in 2006. For other even smaller celestial bodies in our solar system, see p.43.

A celestial body that orbits around a planet is called a **satellite**. The Moon is the Earth's only natural satellite. Planets and satellites do not emit light but reflect light from stars.

In general, a star is much larger than a planet and a planet is larger than a satellite.

✎ Pluto was recognized by International Astronomical Union (IAU) as an important prototype of a new class of Trans-Neptunian Objects in 2008. Such objects were given a new denomination of 'plutoids' (類冥矮行星).

✎ See the website below for the definition of dwarf planet.

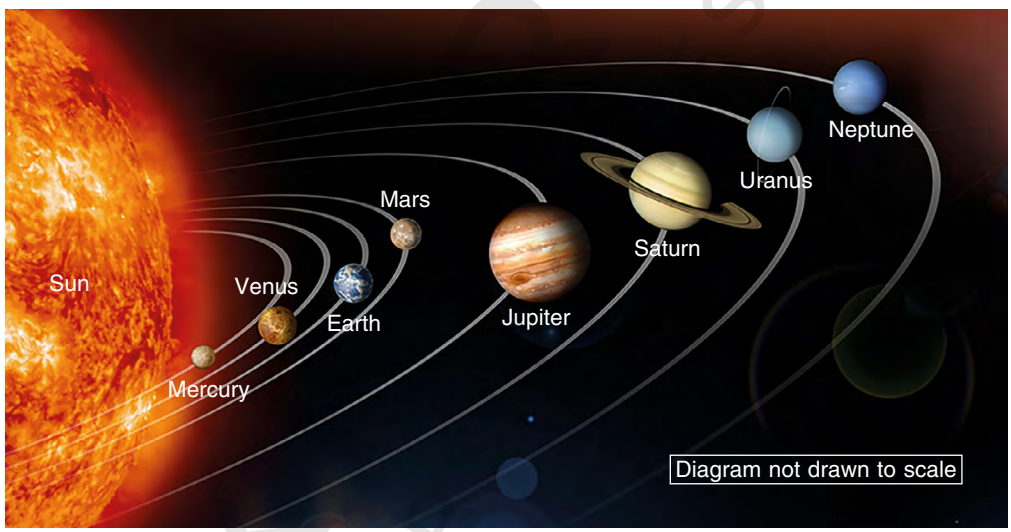


Fig 1.1c *Part of the solar system.*

Know more

Satellites in the solar system

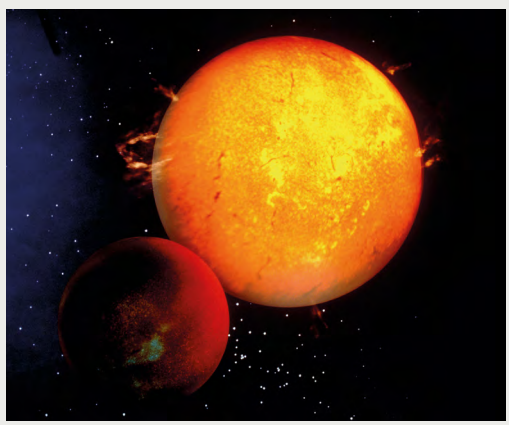
Other planets of our solar system hold different numbers of satellites in their orbits.

Planet	No. of natural satellites
Mercury	0
Venus	0
Earth	1
Mars	2
Jupiter	95
Saturn	146
Uranus	28
Neptune	16

(Data obtained from NASA website in Apr 2024)

Exoplanets

Planets that orbit some other stars rather than our Sun are called *exoplanets*. The first one was confirmed in the 1990s. Since then, thousands have been found. Owing to the advance in observing power and techniques, the number is expected to skyrocket in the future.



An artist's impression of 51 Pegasi b (in the foreground), the first exoplanet discovered orbiting a sun-like star.

planet 行星　solar system 太陽系　dwarf planet 矮行星　satellite 衛星
exoplanet 太陽系外行星

b Star clusters and nebulae

Older stars are usually found in *globular clusters* as shown in Fig 1.1d. Younger stars are usually found in *open clusters* as shown in Fig 1.1e. There are fewer stars in an open cluster generally.

▶ A group of stars bounded by gravity is called a **star cluster** (Fig 1.1d).

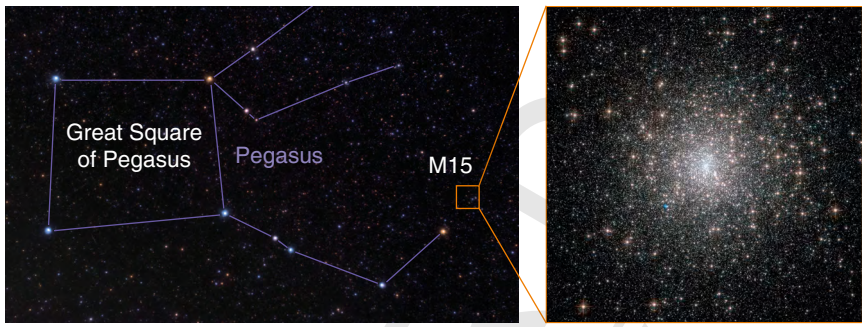
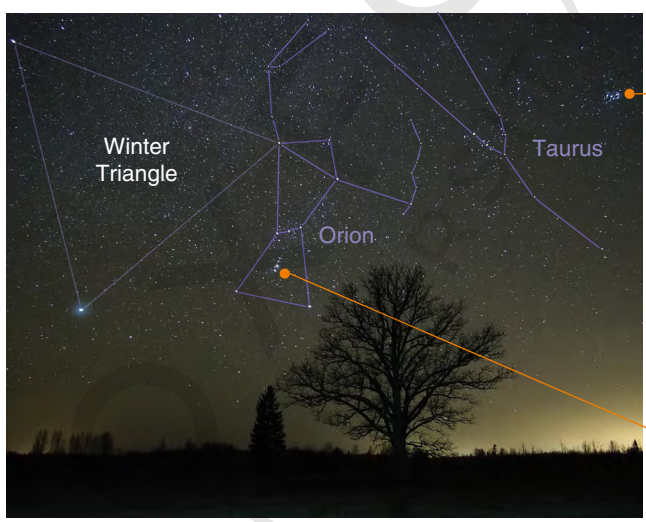


Fig 1.1d *M15, a globular star cluster in the constellation of Pegasus (about 10^{17} m in radius).*

Interstellar means between stars.

▶ **Nebulae** are interstellar clouds of gas and dust. They are the birthplace of stars. The stars in a star cluster are mostly born in the same nebula.

The *Winter Triangle* consisting three bright stars is a prominent sign in the winter night sky of the northern hemisphere.



Pleiades open star cluster
(about 10^{17} m in radius)

Orion Nebula
(about 10^{17} m in radius)

✎ The Orion Nebula (獵戶座大星雲) is in the constellation of Orion, a constellation that can be easily identified in winter night in Hong Kong. The nebula is catalogued as M42 or NGC 1976.

Fig 1.1e *The Pleiades (also known as the Seven Sisters) open star cluster in the constellation of Taurus and the Orion Nebula in the constellation of Orion.*

c Galaxies, galaxy clusters and superclusters

A **galaxy** (Fig 1.1f) is a massive gravitationally bound system which includes stars, star clusters, nebulae, and other matter. Typical galaxies contain 10^7 to 10^{12} stars, all orbiting a common centre.

Spot *P* in Figure 1.1b on p.2 is actually a galaxy. Galaxies can be classified by their shapes. Refer to p.9.

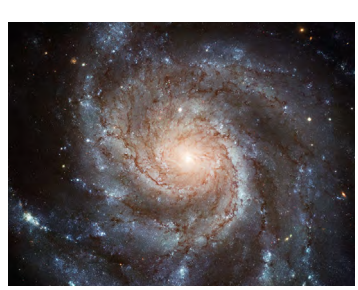


Fig 1.1f *Pinwheel Galaxy.*

star cluster 星團　globular cluster 球狀星團　open cluster 疏散星團　nebula (pl. nebulae) 星雲
Winter Triangle 冬季大三角　galaxy 星系

Our home galaxy is called the *Milky Way Galaxy*. It is a spiral galaxy and is roughly disc-shaped with a bulge at the middle (Fig 1.1g). Our solar system is located far away from its centre. When we look towards the centre of the galaxy from the Earth, we can see the disc of stars as the Milky Way. It is seen as a faint band of light in the sky (Fig 1.1h).

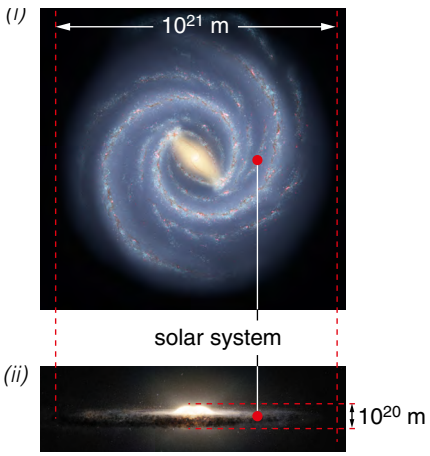


Fig 1.1g *(i) The face-on view and (ii) the edge-on view of the Milky Way Galaxy.*

Fig 1.1h *Panoramic view of the Milky Way.*

A few to a few thousand of galaxies may be bound by gravity to form a **galaxy cluster**. Our home galaxy cluster is called the *Local Group* and spans about 10^{23} m. At an even larger spatial scale, galaxy clusters form groups called **superclusters**. Our home supercluster is called the *Laniakea Supercluster* and spans about 10^{24} m.

Laniakea means immeasurable heaven in Hawaiian. Our home supercluster was once thought to be the *Virgo Supercluster*. Refer to p.9 for details.

The sizes of galaxies, galaxy clusters and superclusters of galaxies are compared in Figure 1.1i.

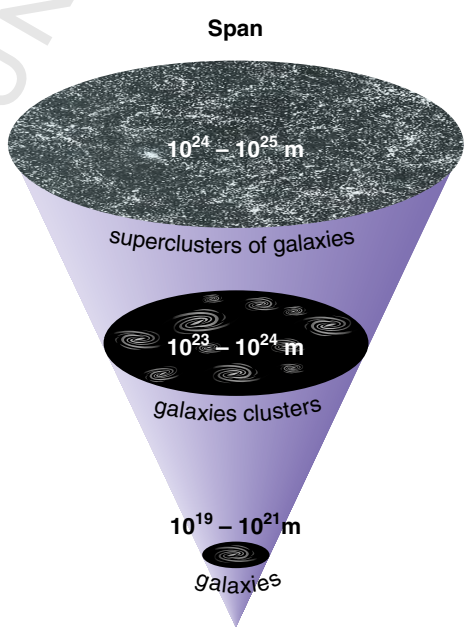


Fig 1.1i *Comparison of size of galaxies, galaxy clusters and superclusters of galaxies.*

Milky Way Galaxy 銀河系 galaxy cluster 星系團 Local Group 本星系羣 supercluster 超星系團
Laniakea Supercluster 拉尼亞凱亞超星系團 Virgo Supercluster 室女座超星系團

DSE goal

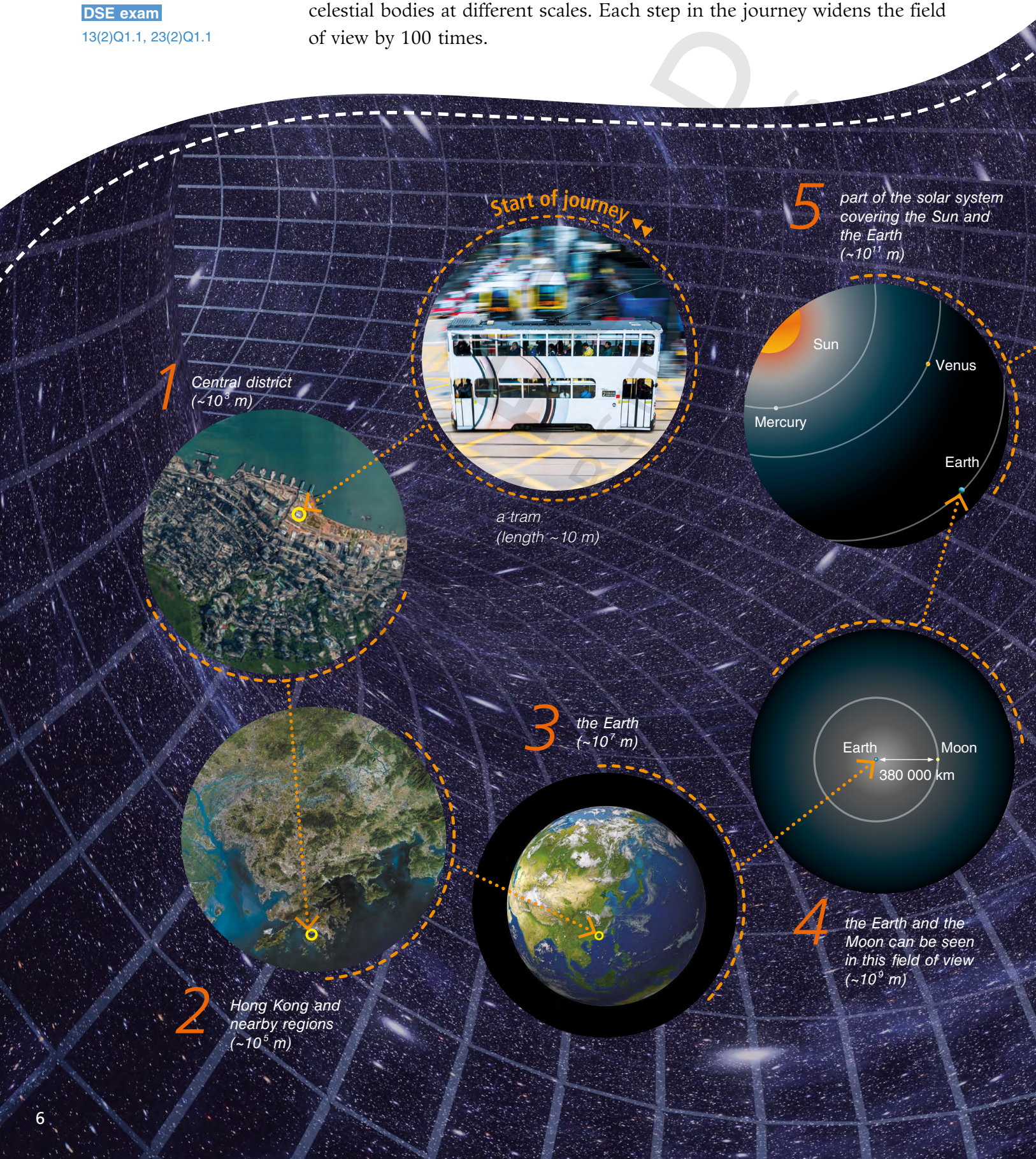
Use the 'Powers of Ten' approach to describe the hierarchy of celestial bodies as seen in different spatial scales.

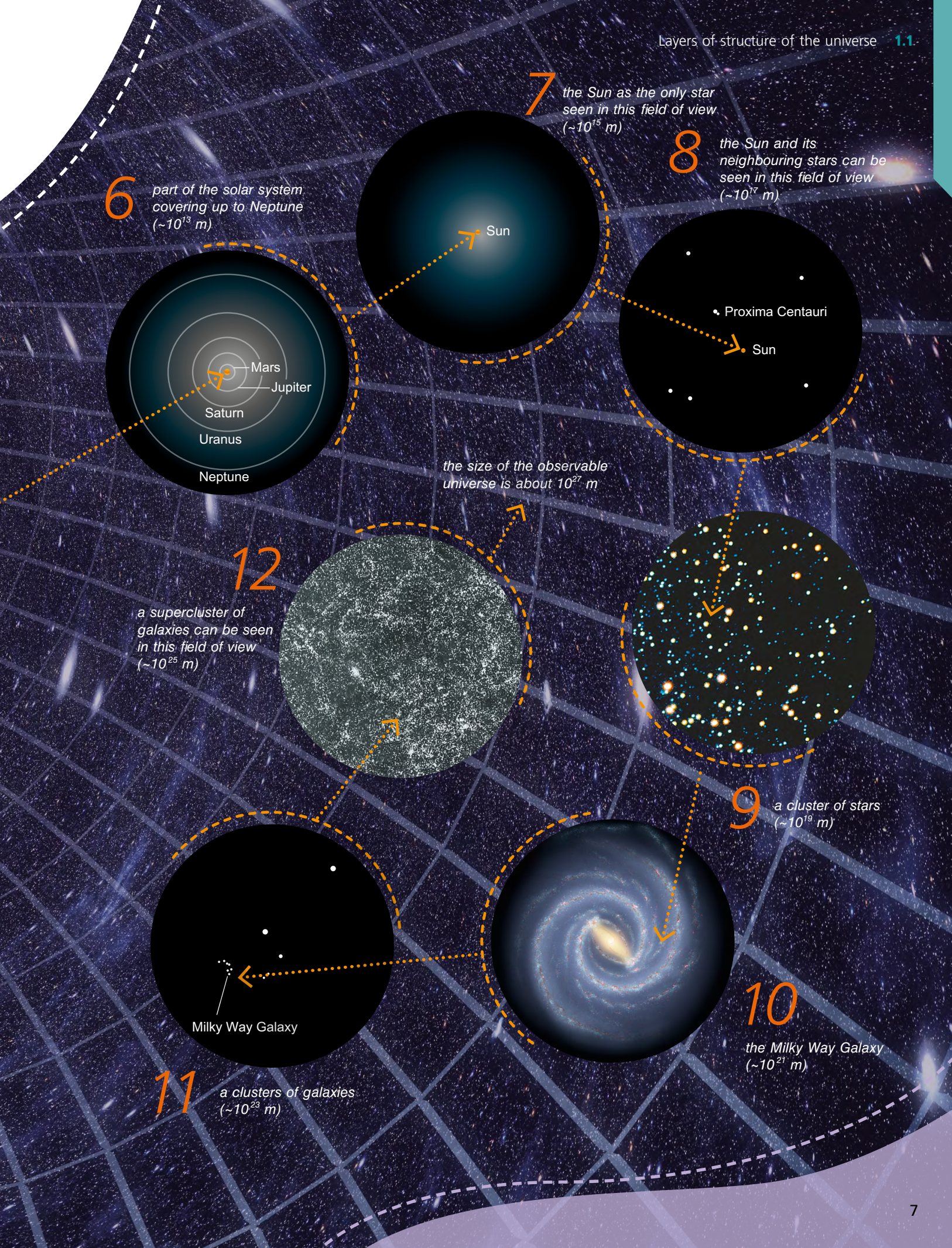
DSE exam

13(2)Q1.1, 23(2)Q1.1

2 A cosmic journey

The universe is so vast. To have a clearer picture of it, let us start with a tram and take a journey through the universe to see the hierarchy of celestial bodies at different scales. Each step in the journey widens the field of view by 100 times.




7
the Sun as the only star seen in this field of view (~10^15 m)
8
the Sun and its neighbouring stars can be seen in this field of view (~10^17 m)
6
part of the solar system covering up to Neptune (~10^13 m)
Sun
Proxima Centauri
Sun
Mars
Jupiter
Saturn
Uranus
Neptune
the size of the observable universe is about 10^27 m
12
a supercluster of galaxies can be seen in this field of view (~10^25 m)
9
a cluster of stars (~10^19 m)
Milky Way Galaxy
10
the Milky Way Galaxy (~10^21 m)
11
a clusters of galaxies (~10^23 m)

Google Forms Microsoft Forms

Quick check 1

→ E-version in OUP Exercise Platform is also available.

1 *True or false*:

(a) There are eight stars in our solar system. (T/F)

(b) Stars generate energy from nuclear fusion. (T/F)

(c) A star cluster consists of many galaxies. (T/F)

(d) A nebula is larger than a galaxy. (T/F)

2 If we want to locate Earth in the universe, how should we write its cosmic address? Arrange the parts of the address below in ascending order of size.

a Earth

b Milky Way

c Laniakea Supercluster

d Local Group

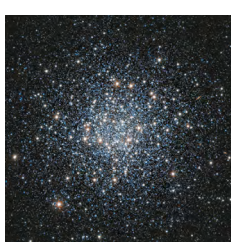
e Solar system

_____, _____, _____, _____, _____

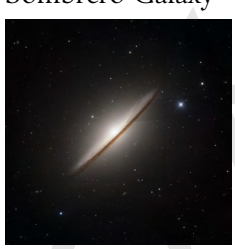
Exercise 1.1

1 Which of the following celestial objects is the biggest in size?

A M55 star cluster



B Sombrero Galaxy



C Jupiter

D The Sun

2 Which of the following statements is/are correct?

(1) Nebulae are the birthplace of stars.

(2) The solar system is inside the Milky Way Galaxy.

(3) A galaxy cluster is larger than a star cluster.

A (1) only

B (1) and (3) only

C (2) and (3) only

D (1), (2) and (3)

***3** Which of the following is **not** contained in the Milky Way (Fig a)?

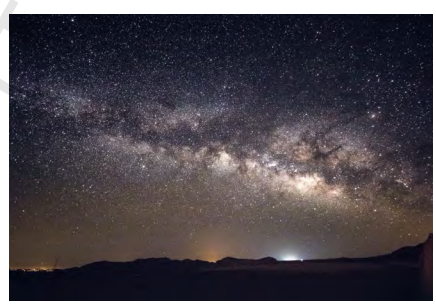


Fig a

A Star clusters

B Galaxy clusters

C Stars

D Solar system

4 Consider the night sky shown below (Fig b). Sirius is a star and Venus is a planet.

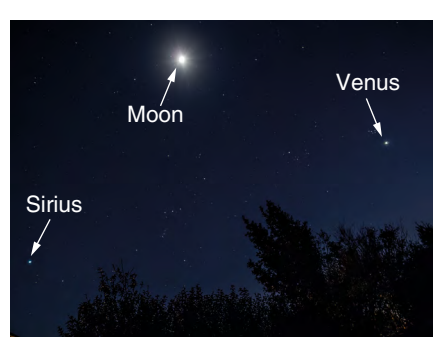


Fig b

(a) Which of them, the Moon, Venus and Sirius, is/are non-luminous object(s)?

(b) Why is the non-luminous object(s) in (a) bright?

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Know more

Types of galaxies

Galaxies have various shapes and sizes. They are classified into three major types according to their shapes: elliptical, spiral and irregular.

Virgo Supercluster

The supercluster of galaxies of which we live in was once considered to be the Virgo Supercluster. However, a study released in 2014 showed that our home supercluster of galaxies, Laniakea Supercluster, is more than 100 times bigger in volume and mass than previously defined. Virgo Supercluster is only a part of it and so is no longer treated as a separate supercluster of galaxies.

Useful link

STSE

Light pollution and astronomical observations

Light pollution is a serious problem in urban areas of Hong Kong. Excessive artificial light causes the night sky to glow and lowers our sensitivity to the stars' dim light. As a result, astronomical observation is rather difficult in Hong Kong. How can we help reduce light pollution? How can we overcome light pollution to carry out astronomical observation?

Hong Kong is one of the heaviest light polluted city on Earth.

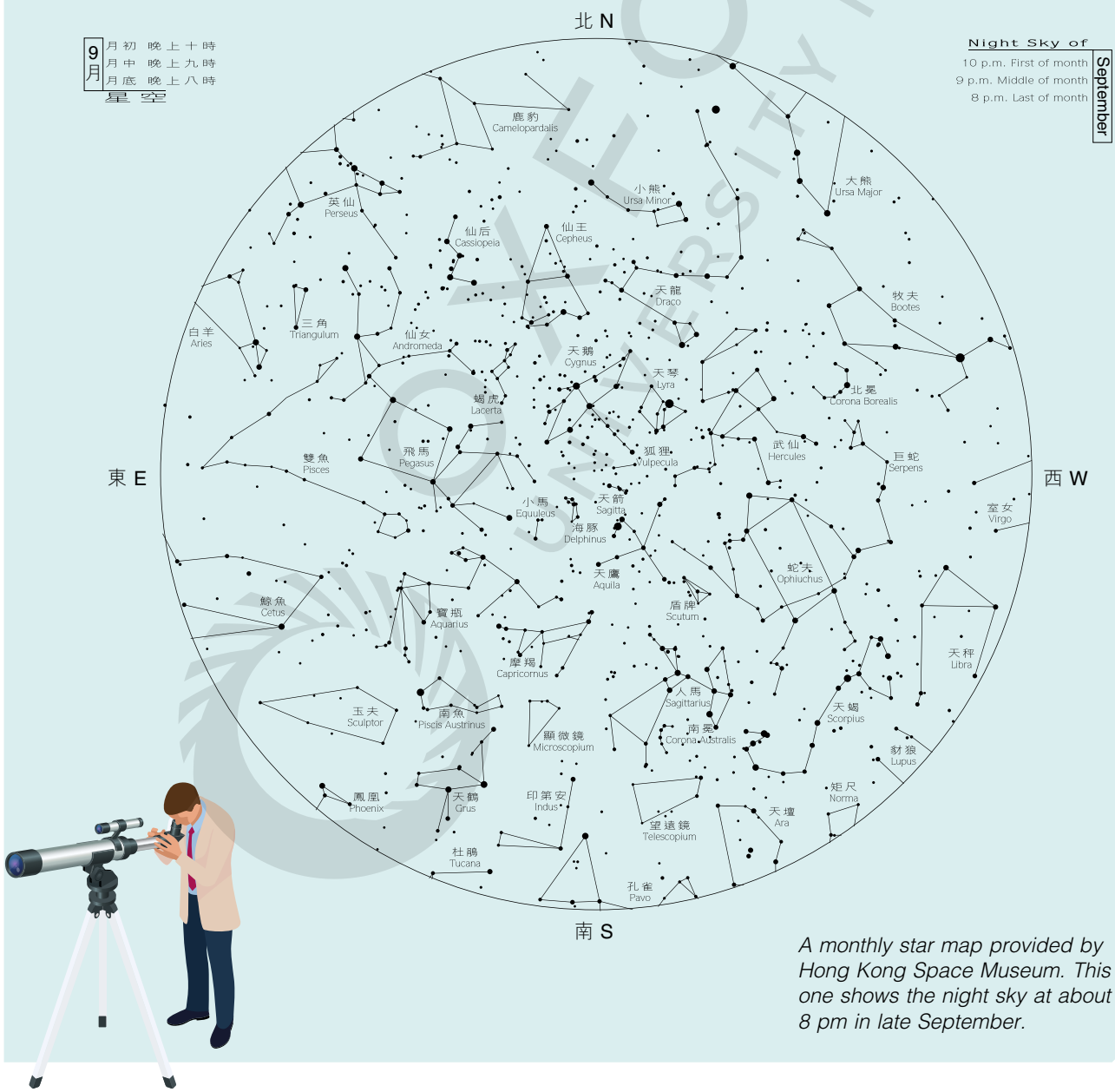
Instruments for stargazing

In a clear night sky, bright planets such as Mercury, Venus and Mars can be observed with the naked eye even on a moonlit night. To observe deep sky objects like nebulae and clusters or planetary features, we need a pair of *binoculars* or a *telescope*.

Binoculars.

Telescope.

For the beginners who cannot identify the stars, it is better to bring *a sky map* (or *star chart*) with you. Sky map is a map showing the location of the stars in the sky. Nowadays, electronic star maps for mobile devices are also available. It is easy to find free app showing star map on the Internet.



A monthly star map provided by Hong Kong Space Museum. This one shows the night sky at about 8 pm in late September.

binoculars 雙筒望遠鏡　telescope 望遠鏡　sky map (or star chart) 星圖

1.2 Describing the spatial scale

The Moon, the closest celestial body to us, is 384 400 km away from the Earth. Light takes 1.28 s to travel from the Moon to the Earth.
The star nearest to the Sun is *Proxima Centauri* (Fig 1.2a). It is 40 208 000 000 000 km away from us. Light from it takes about 4.25 years to reach us!

Proxima Centauri is a red dwarf star located in the constellation of Centaurus (半人馬座).

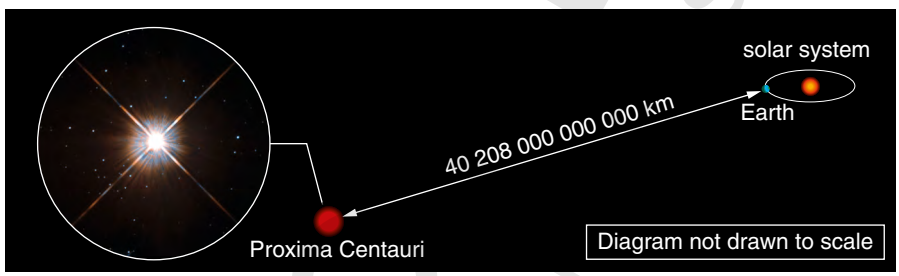


Fig 1.2a *Proxima Centauri.*

The distances involved in astronomical scale are huge. Therefore, it is more convenient to use larger units in such large scale.

DSE goal

Define the basic terminologies such as light year and astronomical unit for describing the spatial scale.

Use parsec (pc) as a unit of distance

DSE exam

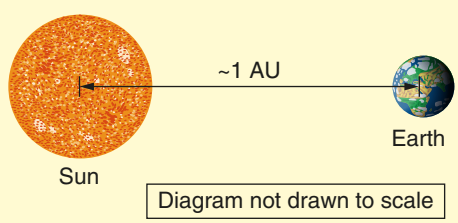
12(2)Q1.3, 14(2)Q1.1, Q1.2, 16(2)Q1.5, Q1.6, 19(2)Q1.1, 20(2)Q.1(a)

1 Units of length

The SI unit of length in physics is the metre (m). However, in astronomy, the unit **astronomical unit** (**AU**) is more often used to express distances **within the solar system**.

The average distance between the Earth and the Sun can be taken as 1 AU.

$$1\ \text{AU} = 1.50 \times 10^{11}\ \text{m}$$



One astronomical unit is exactly equal to 149 597 870 700 m.

Using AU as the unit gives us a more intuitive comparison of the distances relative to the Earth-Sun distance.

The average distances of planets from the Sun are shown below (Fig 1.2b).

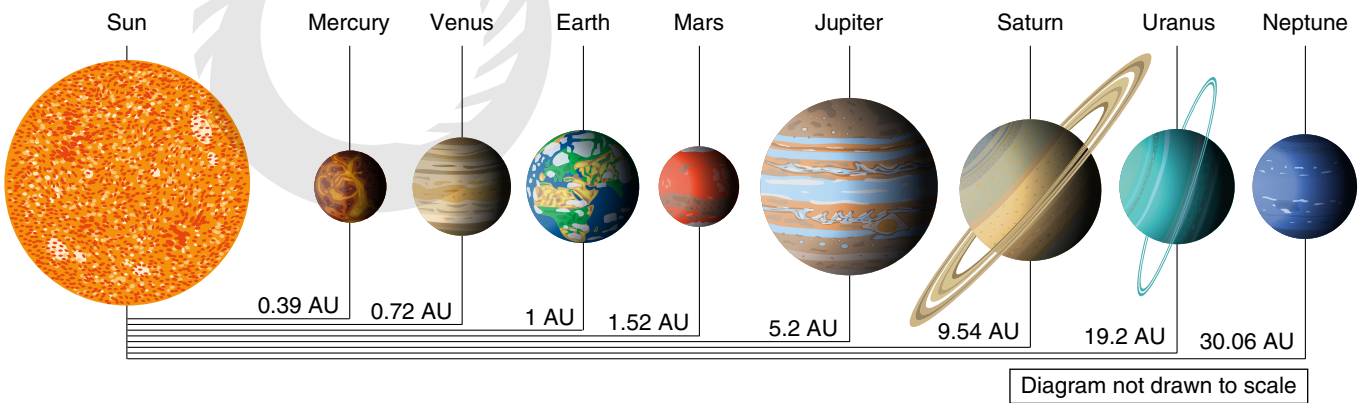


Fig 1.2b *Average distances of planets from the Sun.*

Proxima Centauri 毗鄰星　　astronomical unit (AU) 天文單位

Google Forms Microsoft Forms

Ceres is the only dwarf planet located in the inner solar system. Its radius is just about 0.077 times that of the Earth.

Quick check 2

→ E-version in OUP Exercise Platform is also available.

1 Complete the table below.

	Celestial body	Average distance from the Sun	
		in AU	in m
(a)	Venus	0.72	
(b)	Ceres		4.14×10^{11}
(c)	Neptune	30.06	

Caution

The light year is a unit of length, not time.

For distance in **interstellar or even intergalactic scale**, another larger unit called **light year** (**ly**) is often used.

Light from Proxima Centauri needs 4.25 years to reach the Earth. This means the distance between Proxima Centauri and the Earth is 4.25 ly.

One light year (ly) is the distance travelled by light in a vacuum in one year.

The speed of light is exactly equal to 299 792 458 m s^{-1}.

Since light travels at a speed of 3.00×10^8 m s^{-1} in a vacuum, one light year can be converted to metres by:

1 year = 365.25 days

$$1 \text{ ly} = 3.00 \times 10^8 \times (365.25 \times 24 \times 60 \times 60) \text{ m}$$

Ss may think that one year exactly equals 365 days. Remind them that one year cannot be presented by a whole number of calendar days. An extra day is inserted into the calendar to keep the calendar year synchronized with the Earth's orbital motion around the Sun. A leap year is divisible by four. But for the years ending in -00, they will not receive the extra day unless they are divisible by 400.

$$\mathbf{1 \text{ ly} = 9.46 \times 10^{15} \text{ m}}$$

If we know the distance of a celestial object from us in ly, we can tell the time for its light to reach us immediately.

The average distances in light years from the Earth to some celestial objects are shown below (Fig 1.2c).

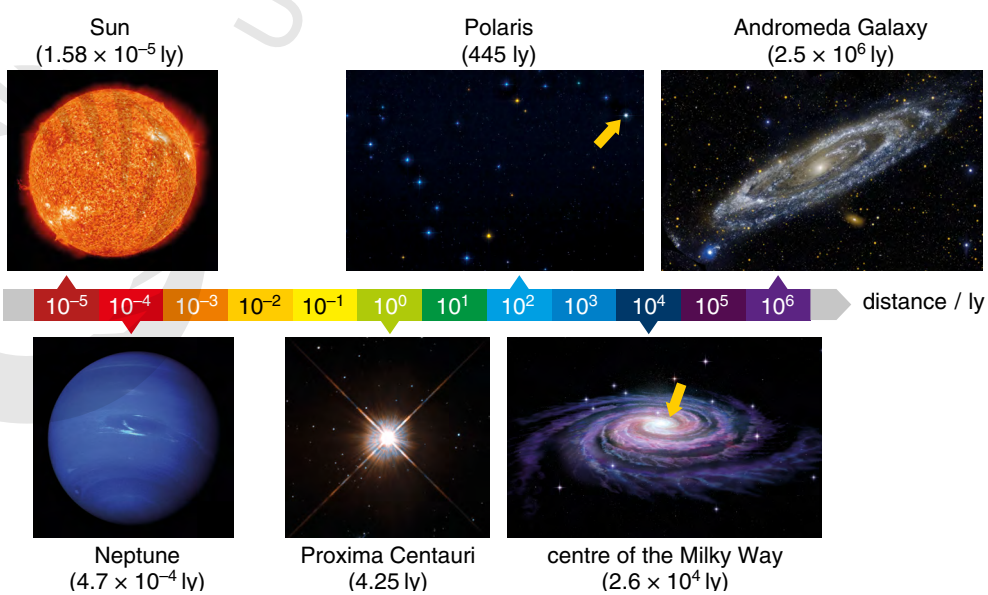


Fig 1.2c *The average distances from the Earth to some celestial objects.*

light year (ly) 光年

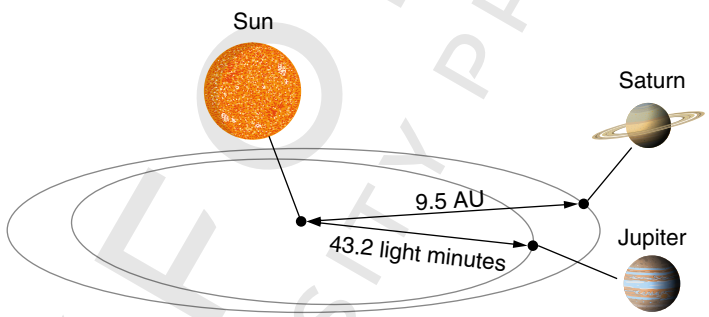
Apart from light year, **parsec** (**pc**) is also a common unit used in expressing interstellar or intergalactic distances.

Cross-link
More information about parsec will be given in Ch 3 (p.100).

$$\mathbf{1\ pc = 3.09 \times 10^{16}\ m = 3.26\ ly = 206\ 265\ AU}$$

Example 1 Distances of different planets from the Sun

Jupiter and Saturn are two large planets composed mostly of gases.



Light hour, light minute and light second are used to express shorter distances in astronomy. One light second is the distance travelled by light in a vacuum in a second.

(a) Light minute is another unit of length. A light minute is the distance travelled by light in a vacuum in one minute. Jupiter is about 43.2 light minutes from the Sun.
  (i) How long does it take for light to travel from the Sun to Jupiter?
  (ii) Express the distance between the Sun and Jupiter in metres.

(b) The distance between the Sun and Saturn is about 9.5 AU.
  (i) Express the distance in metres.
  (ii) Estimate the time needed for the light from the Sun to reach Saturn. Give your answer in seconds.

Take the speed of light $c = 3.00 \times 10^8$ m s^{-1}.

Solution

(a) (i) 43.2 minutes

(ii) 43.2 light minutes $= 43.2 \times (3.00 \times 10^8) \times 60$
$= 7.78 \times 10^{11}$ m

(b) (i) $9.5\ \text{AU} = 9.5 \times 1.50 \times 10^{11}$ m
$= 1.425 \times 10^{12}$ m
$\approx 1.43 \times 10^{12}$ m

(ii) Time needed $= \dfrac{1.425 \times 10^{12}}{3.00 \times 10^8} = 4750$ s

parsec (pc) 秒差距

Google Forms Microsoft Forms

Quick check 3

→ E-version in OUP Exercise Platform is also available.

Unless otherwise specified, take the speed of light $c = 3.00 \times 10^8$ m s^{-1}.

1 Which of the following is **not** a unit of length?

A Parsec B Astronomical unit
C Year D Light year

2 Complete the unit conversions below.

(a)	2.6×10^4 ly	m
(b)	7.8×10^5 pc	m
(c)	8.6 ly	pc

Learn by practice 1

Unless otherwise specified, take the speed of light $c = 3.00 \times 10^8$ m s^{-1}.

1 Arrange the following celestial bodies in ascending order of distance from the Earth. The figure in the brackets shows the distance of the object from the Earth.

(a) Sun (b) Markab (40.9 pc)
(c) Scheat (196.0 ly) (d) Uranus (19 AU)

2 The distance between the Earth and the Moon is about 1.28 light seconds.

(a) How long does it take for the light to reach the Earth from the Moon?

(b) Express the distance between the Earth and the Moon in metres.

☆ HKDSE 2016 Paper 2 Q1.6, 2020 Paper 2 Q.1

→ This video demonstrates how this question is solved.

Video

DSE practice 1 | Supernova

A supernova is a massive explosion of a dying star. The explosions are extremely bright and powerful. Supernova SN1604, also called Kepler's supernova, was first observed in October 1604 at a distance of 6.1 kpc from the Earth (Fig a).

(a) Estimate how long ago the explosion of SN1604 took place before its light reached the Earth. (2 marks)

(b) In 1054, another supernova SN1054 which is at 6500 light years away from the Earth was observed. How many years did SN1604 explode before SN1054? (1 mark)

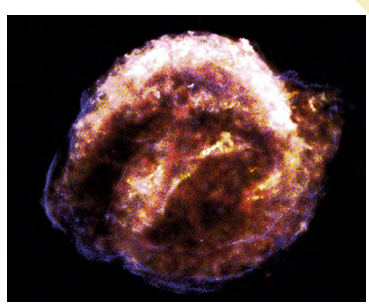


Fig a *The remnant of Kepler's supernova.*

Solution

(a) Distance 6.1 kpc = 6100×3.26 ly 1M
= 19 900 ly

The explosion happened 19 900 years ago before the light reached the Earth. 1A

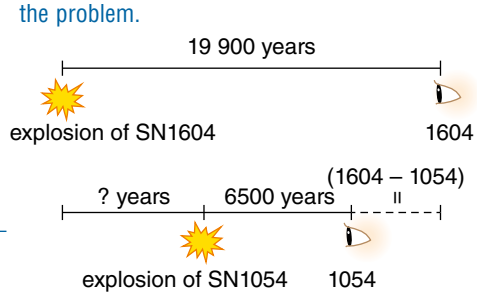
(b) The explosion of SN1054 took place 6500 years before it was observed.

19 900 – 6500 – (1604 – 1054)

= 12 850 years

SN1604 exploded 12 850 years before SN1054. 1A

Timelines can be drawn to help solve the problem.



▶ Revision 1 Q9 (p.39)

Exercise 1.2

Unless otherwise specified, take the speed of light $c = 3.00 \times 10^8$ m s^{-1}, 1 AU = 1.50×10^{11} m, 1 ly = 9.46×10^{15} m, 1 pc = 3.09×10^{16} m = 3.26 ly = 206 265 AU.

1 Light year is a unit of

A time. B length.
C speed. D angle.

2 Arrange the following lengths in ascending order.

(1) 2 pc
(2) 5 ly
(3) 8 AU

A (1), (2), (3) B (2), (3), (1)
C (3), (1), (2) D (3), (2), (1)

★**3** Crab Nebula is expanding outward at a steady rate of 1500 km s^{-1}. It spans about 11 ly now (Fig a). How long will it take for its span to become 12 ly?

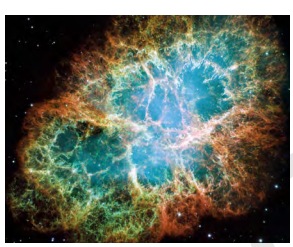


Fig a

A 100 years B 200 years
C 2200 years D 2400 years

★**4** The galactic disc of Milky Way Galaxy is 105 700 ly across and 1000 ly thick. Astronomers estimate that the disc contains about 200 billion stars. Assume the stars are evenly distributed through the galaxy. What is the average separation of the stars? (Hint: volume of a cylinder = $\pi r^2 h$)

A 43.9 ly B 3.53 ly
C 175 ly D 5.60 ly

★**5** The size of a proton is of the order of 10^{-15} m. The size of the Virgo Supercluster is of 33×10^6 pc. What is the volume ratio of a proton to the Virgo Supercluster?

A 10^{-39} B 10^{-68}
C 10^{-78} D 10^{-117}

6 Our home galaxy Milky Way Galaxy has a radius of about 75 000 ly.

(a) How long does it take for light to travel across Milky Way Galaxy?
(b) Express the diameter of Milky Way Galaxy in parsec.

7 Voyager 1, the first spacecraft to cross into interstellar space, is still on its journey (Fig b). It was at a distance of 163 AU from the Earth in January 2024.

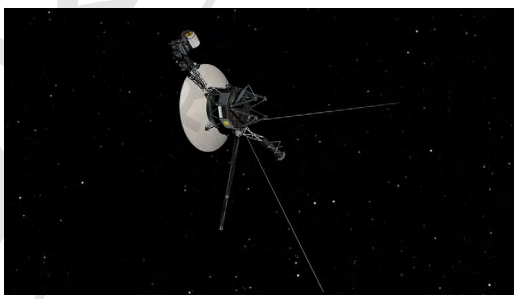


Fig b

(a) Express 163 AU in light years.
(b) Estimate the elapsed time in seconds for the radio signals sent by Voyager 1 to reach the Earth at that time.

★**8** Mars is a planet in the solar system (Fig c).

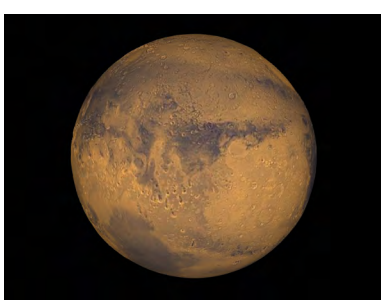


Fig c

(a) Mars is at 1.52 AU from the Sun. Express the distance in (i) metres and (ii) light minutes.
(b) How long does it take for sunlight to reach Mars?
(c) Mars takes 687 days to complete one revolution around the Sun. What is its average orbital speed in km s^{-1}?

★**9** The Sun is about 8000 pc from the centre of the Milky Way Galaxy. It takes 225 million years for the Sun to revolve around the galactic centre once. Suppose the orbit of the Sun around the galactic centre is circular. Estimate the linear speed of the Sun in km s^{-1}.

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1.3 Models of planetary motion

Figure 1.3a is a photo of the night sky using long exposure time. From the star trails, we can easily spot that the stars have moved across the sky. Ancient astronomers did not have cameras, but they did make remarkable discoveries about the sky.

Fig 1.3a *Long-exposure photo showing star trails over Clear Water Bay Country Park.*

1 Observation of ancient astronomers

a Motion of stars

After generations of careful observations, ancient people concluded that **every night** the stars move as follows:

1 Stars move slowly across the sky **from the east to the west**.

2 Almost all stars appear to be stationary with respect to one another when they move across the sky. That is, they all move together as a group at the **same rate** across the sky (Fig 1.3b).

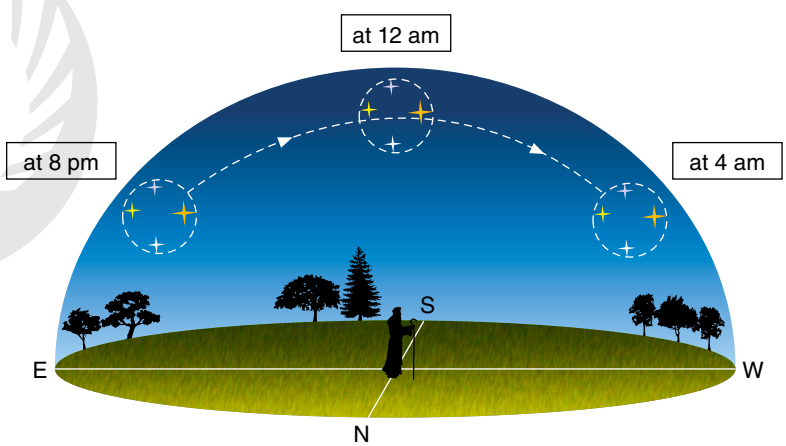


Fig 1.3b *Stars move together as a group across the sky.*

Know more

Celestial sphere

Ancient astronomers regarded the sky as an imaginary sphere called a *celestial sphere*. Do you know how this sphere works?

▸ p.44

celestial sphere 天球

Simulation

→ This simulation shows the prograde motion and retrograde motion of planets. It is available on OUP web in PC version (.exe) only.

b Retrograde motion of planets

When ancient astronomers observed the sky **over months**, they found a few extraordinary 'stars' that do not move in sync with other stars. Therefore, these 'stars' were given the name 'planets'. The planets seen by the ancients with naked eye are Mercury, Venus, Mars, Jupiter and Saturn.

The name 'planet' is derived from the Greek word 'planetes' which means 'wanderer'. ▶

✎ These five planets can be discerned with the naked eye without much difficulty. Therefore, they were the only planets known to the ancient astronomers.

In most of the time, the planets **shift slightly eastwards** against the background stars **from night to night** (Fig 1.3c). Such **eastward** motion of planets is called **prograde motion**.

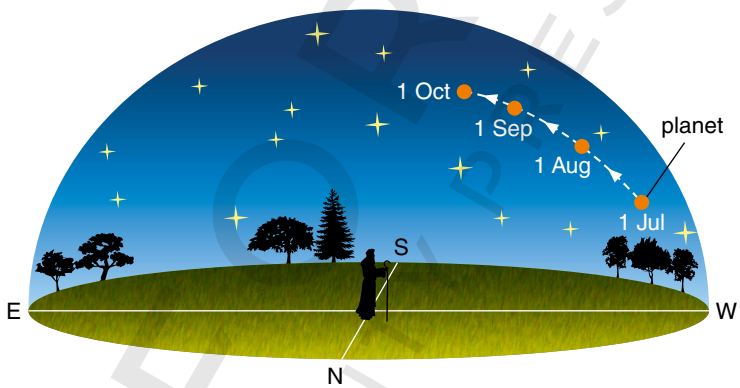


Fig 1.3c *The path of a planet against the background stars can be traced over a few months.*

Planets do not move at a constant rate and always eastwards. Sometimes they move westwards for a few weeks or a few months, then resume to move eastwards again (Fig 1.3d). This **westward** motion is called **retrograde motion**.

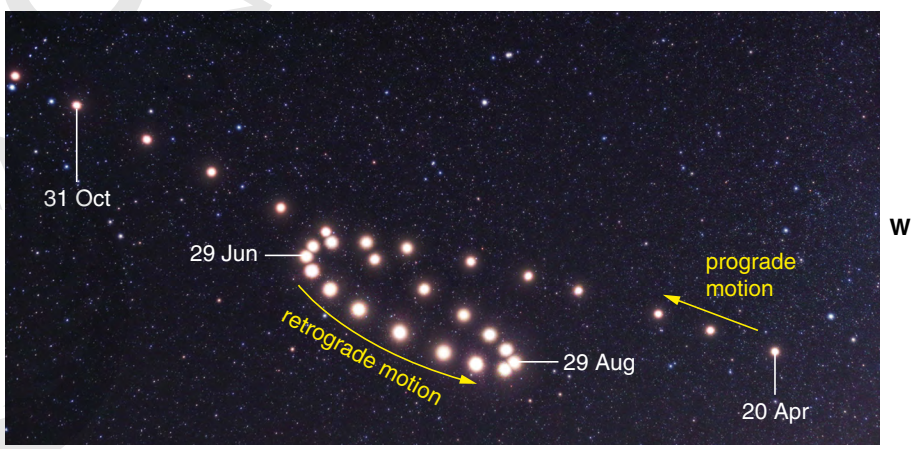


Fig 1.3d *A composite photo showing the retrograde motion of Mars in 2018.*

Mars grows brighter during retrograde motion. The retrograde motion and the changes in brightness of planets puzzled ancient astronomers. ▶

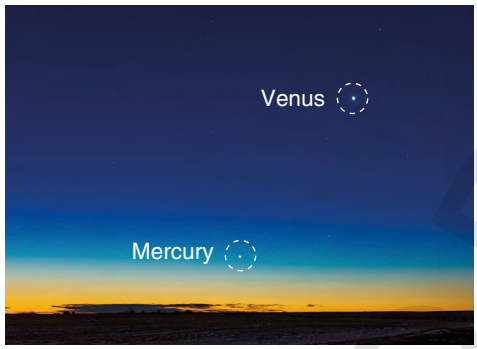
Retrograde motion is the apparent westward motion of a planet as seen from the Earth against the background stars.

prograde motion 順行　　retrograde motion 逆行

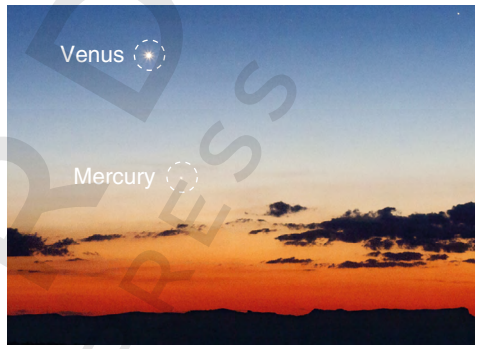
C Motion of Mercury and Venus

Ancient astronomers also observed that **Mercury and Venus are never seen far from the Sun**. They either appear above the eastern horizon just before sunrise, or above the western horizon just after sunset (Fig 1.3e). The ancients therefore called them as morning 'stars' and evening 'stars'.

The ancients once regarded Mercury and Venus appearing in the two time intervals as different 'stars'. They were called 'Apollo' and 'Phosphorus' as morning 'stars', and 'Hermes' and 'Hesperus' as evening 'stars'.



(i) Mercury and Venus as morning 'stars' above the eastern horizon at dawn.



(ii) Mercury and Venus as evening 'stars' above the western horizon at dusk.

Fig 1.3e *Mercury and Venus are always near the Sun, which is just below the horizon.*

Quick check 4

→ E-version in OUP Exercise Platform is also available.

Google Forms | Microsoft Forms

1 *True or false*:

(a) Stars rise from the west and set at the east every night. (T/F)

(b) Stars appear to be stationary with respect to one another and move slowly across the sky. (T/F)

(c) Planets drift slowly eastwards against the background stars. Occasionally, they move backwards, and this westward motion is called retrograde motion. (T/F)

(d) Mercury and Venus are seen far from the Sun. (T/F)

2 When observed from the Earth, which of the following planets only appear(s) either above the eastern horizon just before sunrise, or above the western horizon just after sunset? Put a tick in the corresponding box(es).

☐ Mercury ☐ Jupiter ☐ Venus ☐ Mars

Know more

Findings of ancient astronomers about retrograde motion

With the naked eye, ancient astronomers were still able to note the followings.

1. Each planet appears to move slower when it reverses its moving direction.
2. The planets look brighter during retrograde motion.
3. Each planet has its own period, e.g. the retrograde motion of Mars happens every 26 months.
4. The durations of retrograde motion of different planets vary, from a few weeks to a few months.
5. The path showing the retrograde motion of a planet can either be a loop or a zigzag pattern.

A zigzag path of Mars was traced from July 2005 to February 2006.

Flipped classroom

Watch a video and answer the questions.

DSE goal

Use the geocentric model to explain the motion of planets on the celestial sphere.

DSE exam

17(2)Q1.1, 20(2)Q1.6

See more about the geocentric model of the ancient Greeks on p.44.

Simulation

→ This simulation shows how planets and the Sun move in Ptolemy's geocentric model. It is available on OUP web in PC version (.exe) only.

2 The geocentric model

The retrograde motion and the changes in brightness of the planets puzzled ancient astronomers for a long time. They tried to construct models to account for the phenomena. A **geocentric model** was suggested by ancient Greek astronomers and widely accepted by many astronomers.

The most highly developed geocentric model was the one proposed by *Claudius Ptolemy* at around 140 AD. The Ptolemaic model could reasonably explain the apparent motions of the planets and predict the positions of the planets with certain accuracy.

Ts may state that scientists often build models to explain what they observe. The more observations a model can explain, the more powerful the model is.

a The Ptolemaic model

Figure 1.3f shows a simplified form of Ptolemy's geocentric model.

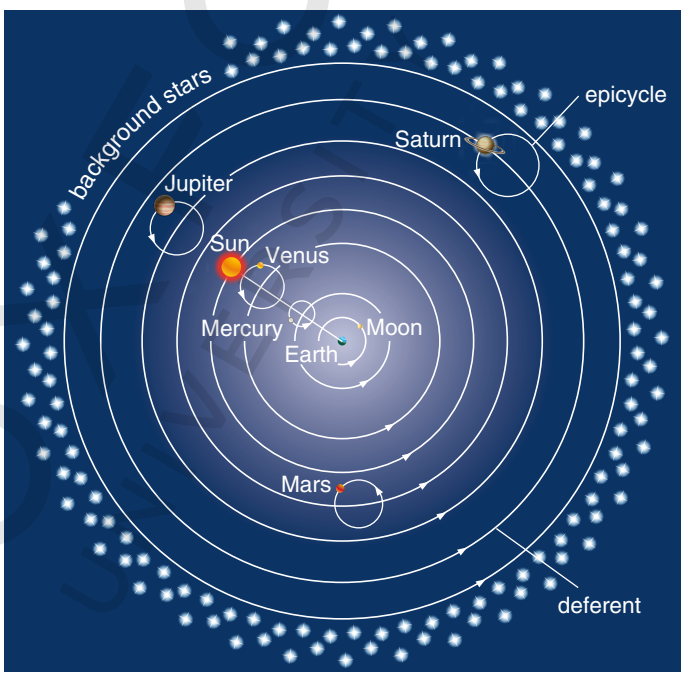


Fig 1.3f *A simplified form of Ptolemy's geocentric model.*

Below are some features of this model:

In fact, the Earth is slightly off-centre in Ptolemy's geocentric model.

1. The Earth is stationary at the centre of the universe and the background stars, which are fixed on the outermost sphere, move around the Earth. Ts may remind Ss that the outermost sphere is the celestial sphere.
2. The Sun and the Moon move around the Earth in circular orbits.
3. Each planet moves in a small circle called an **epicycle**, whose center moves around the Earth along a large circle called a **deferent**.
4. The centres of the epicycles of Mercury and Venus always lie on a line joining the Earth and the Sun.

geocentric model 地心模型　Claudius Ptolemy 托勒密　epicycle 本輪　deferent 均輪

i Explaining the retrograde motion

The retrograde motion of planets can be explained by the combined effect of epicycle and deferent. The planets would eventually move along a spiral-like orbital path, so they have prograde and retrograde motion as seen from the Earth (Fig 1.3g).

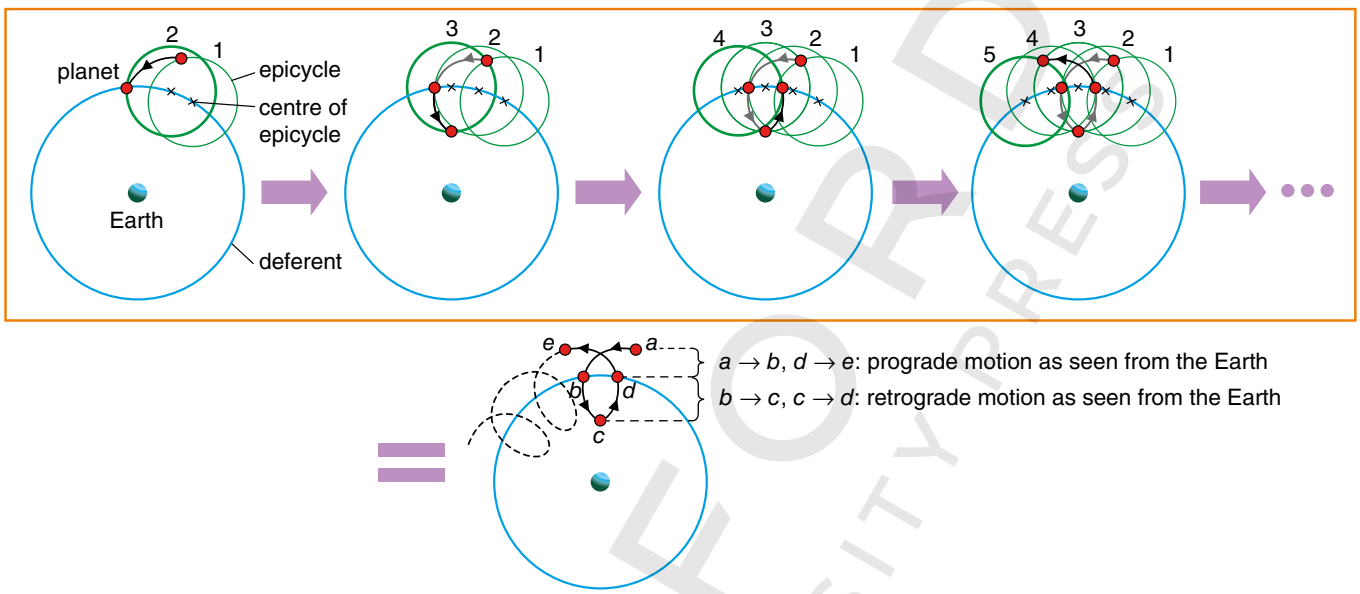


Fig 1.3g *Explaining retrograde motion using epicycle and deferent.*

ii Explaining the changes in brightness

The Ptolemaic model also accounted for the changing brightness of planets. From Figure 1.3g, we could see that when the planet retrogrades, it moves closer to the Earth. Hence, the planet appears brighter to us. The brightness of planet changes because our distance to the planet varies.

iii Explaining the motions of Mercury and Venus

In the Ptolemaic model, the centres of the epicycles of Mercury and Venus always **lie on a straight line joining the Earth and the Sun** (Fig 1.3h). This explains why **these two planets never move far away from the Sun**.

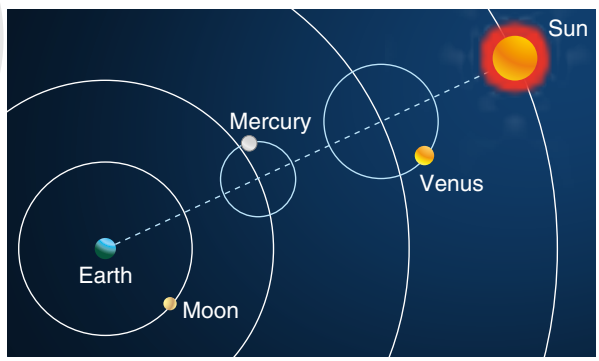


Fig 1.3h *The centres of the epicycles of Mercury and Venus were fixed along a line between the Sun and the Earth.*

Nature of science

From observation to hypothesis

Scientists often make *hypotheses* to their observations. Do you know what a hypothesis is?

▸ p.42

hypothesis (pl. hypotheses) 假說

DSE goal

Use the heliocentric model to explain the motion of planets on the celestial sphere.

DSE exam

15(2)Q1.5, Q1.6

When there are two models with similar explanatory power, the simpler one will be regarded as a better model.

3 The heliocentric model

Ptolemy's geocentric model was generally accepted in the western world for nearly 1500 years. However, as time went by, there were discrepancies between predictions and observations. Astronomers tried to refine the model by adding many circles, but this made the model very complicated. People began to pursue a better model.

a The Copernican model

In the 16th century, *Nicolaus Copernicus* proposed the **heliocentric model** which put the Sun but not the Earth at the centre of the universe with the planets revolving around it (Fig 1.3i). This was a revolutionary concept. Compared with the Ptolemaic model, the Copernican model could provide simpler and more natural explanations for the planetary motions.

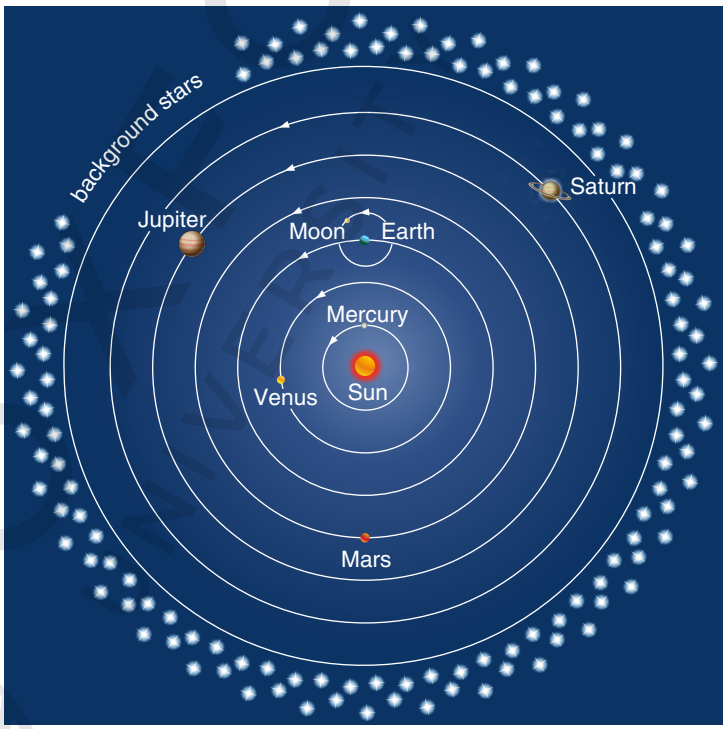


Fig 1.3i *Copernicus' heliocentric model.*

The model has the following features.

1 The Sun is stationary at the centre of the universe. The Earth and other planets move around the Sun in circular orbits. The background stars are fixed on the outermost sphere.

2 The orbital speed of the planets decreases with increasing distance from the Sun. This means Mercury moves the fastest in its orbit while Saturn is the slowest.

This explains the daily motion of all celestial objects moving across the sky from the east to the west ▶

3 The Earth rotates on its own axis from West to East.

4 Only the Moon orbits around the Earth.

Nicolaus Copernicus 哥白尼　　heliocentric model 日心模型

Simulation

→ This simulation shows how Copernican model explains the retrograde motion of planets. It is available on OUP web in PC version (.exe) only.

i Explaining the retrograde motion

In the Copernican model, the Earth in the inner orbit moves faster than Mars does. When the Earth happens to **overtake Mars**, **Mars appears to move backwards (i.e. westwards) in the sky** as observed from the Earth (Fig 1.3j). This results in the retrograde motion of the planet.

Since the planetary orbits do not lie in precisely the same plane, a planet does not resume its eastward motion in precisely the same path it followed earlier.

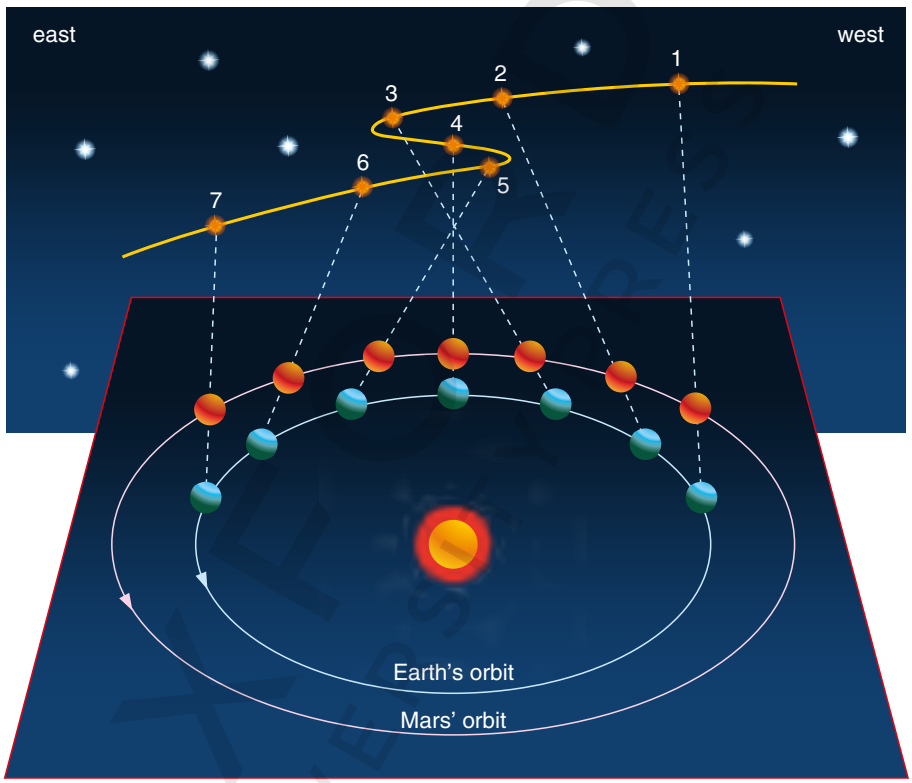


Fig 1.3j *Explaining retrograde motion using the Copernican model. The yellow path shows the apparent path of Mars as observed from the Earth.*

Contrary to the Ptolemaic model, planets in the Copernican model do not really move backwards in their orbits.

ii Explaining the changes in brightness

From Figure 1.3j, we can see that Mars is not always at the same distance from the Earth. Therefore, Mars appears brighter when it is closer to the Earth (Fig 1.3k). Similarly, the brightness of other planets varies periodically.

In fact, the brightness of a planet is also affected by the phase of the planet.

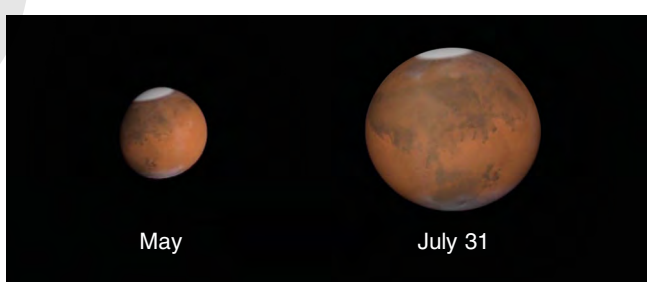


Fig 1.3k *Mars appears brighter and bigger when it is at the closest position to the Earth on 31 July 2018.*

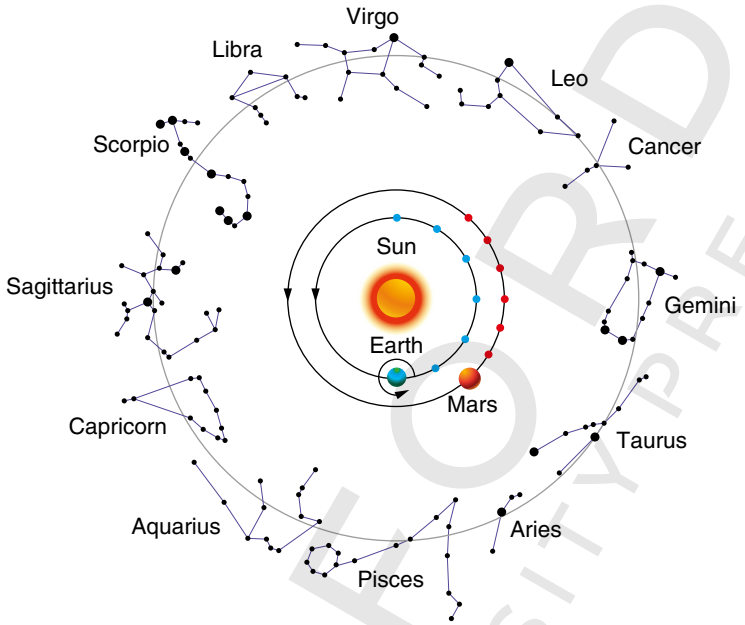
☆ HKDSE 2015 Paper 2 Q1.5, Q1.6

→ This video demonstrates how this question is solved.

Video

DSE practice 2 | The motions of the Earth and Mars

The Earth and Mars are at the positions shown below. The blue and red dots indicate their positions in each of the next six consecutive months.



(a) Suppose the Earth is at the position shown in the figure.

(i) Which constellation can be seen to rise at the eastern horizon at sunset? (1 mark)

(ii) Can Virgo be seen in the night sky? Explain. (2 marks)

(b) After how many months will Mars appear to be the brightest in the sky? Explain. (2 marks)

(c) In the next week after Mars is the brightest, does it appear to move eastwards or westwards against the background stars? (1 mark)

Solution

(a) (i) Pisces. 1A

At sunset, Pisces is on the opposite side of the Earth.

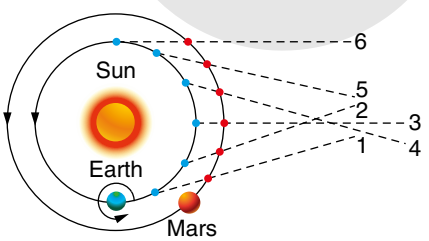
(ii) No. 1A

The Sun passes in front of Virgo and Virgo cannot be seen at night. 1A

(b) Mars will appear to be the brightest after three months. 1A

This is because it is closest to the Earth at that time. 1A

(c) Mars appears to move westwards against the background stars. 1A



Common mistake ⚠

When the Earth lines up with the Sun and Mars, Mars is closest to the Earth and appears the brightest. Students may not realize that Mars retrogrades at that time and will continue to retrograde for another month.

▶ Revision 1 Q10 (p.39)

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iii Explaining the motions of Mercury and Venus

Mercury and Venus, whose orbits are inside the orbit of the Earth, are called *inferior planets*. The planets outside the Earth's orbit are called *superior planets*.

▶ The heliocentric model also explains why Mercury and Venus always appear close to the Sun, as seen from the Earth. This is because **their orbits are inside the Earth's orbit**.

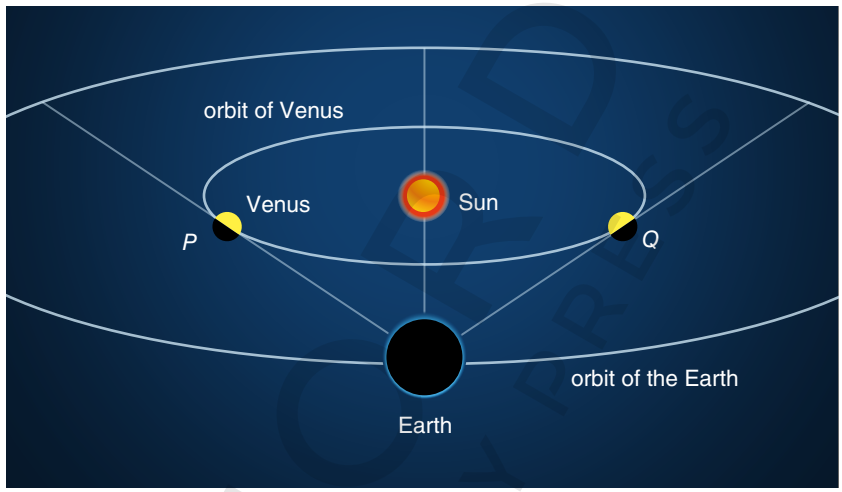

Fig 1.3l *The orbit of Venus lies within that of the Earth.*

From Figure 1.3l, Venus is furthest away from the Sun at *P* and *Q* as seen from the Earth. Therefore, Venus only swings to and fro to either side of the Sun and it is never far from the Sun.

DSE goal

Compare the heliocentric model with the geocentric model in explaining the motion of planets on the celestial sphere.

DSE exam

13(2)Q1.2, 18(2)Q1.3, 19(2)Q1.2, 23(2)Q1.2

4 Limitations of the Ptolemaic and Copernican models

Both the Ptolemaic and Copernican models have limitations.

For the Ptolemaic model:

1 It did not tell why the epicycles of Mercury and Venus had to bind with the Sun while other planets could be treated independently.

2 To eliminate the discrepancies between predictions and observations, astronomers had added many circles to refine the model and made the model very complicated.

For the Copernican model:

In the Copernican model, planet orbits were still modelled as perfect circles. However, the real orbits are elliptical in shapes. This discrepancy is the major reason for the differences between the observations and the predictions.

▶ 1 It was not more accurate than the Ptolemaic model in predicting planetary positions. ✎ Copernican model can be used to find the planet's period and its distance from the Sun.

2 To improve its accuracy, Copernicus still used minor epicycles. This made his model equally complicated but not more accurate than the Ptolemaic model. ✎ Although Copernicus was not completely correct, his idea that the Earth is not at the centre of the universe is correct. This proved to be critical in the development of our understanding of the solar system. Today we call his idea the Copernican revolution.

inferior planet 內行星　　superior planet 外行星

Table 1.3a compares the Ptolemaic and Copernican models.

Model	Ptolemaic	Copernican
Type	geocentric (Earth-centred)	heliocentric (Sun-centred)
Use of circular orbits	yes	yes
Background stars	rotate around the Earth in the outermost sphere	remain stationary in the outermost sphere
Use of epicycle	to explain retrograde motion	to increase accuracy
Explaining retrograde motion	combined effect of epicycle and deferent	relative motion of the Earth and planets due to different orbital speeds of planets
Explaining brightness of planets	Distances between the Earth and planets vary as the planets move along the epicycle.	Distances between the Earth and planets vary due to their different orbital speeds.
Explaining motions of Mercury and Venus	The centres of the epicycles of Mercury and Venus were assumed to lie on a straight line connecting the Sun and the Earth.	The orbits of Mercury and Venus are inside the Earth's orbit.
Accuracy of prediction	☺ relatively more accurate on predicting planets' positions	☹ not more accurate than the Ptolemaic model even after adding epicycles
Complexity	☹ too complicated	☺ simpler

Table 1.3a *Comparisons of the Ptolemaic and Copernican models.*

In Ptolemaic model, the orbital distances of the planets are arbitrary.

The Copernican model was not supported by many, especially those in the Roman Catholic Church initially. The situation changed when *Galileo Galilei* made his discoveries which disproved the Ptolemaic model.

Quick check 5

→ E-version in OUP Exercise Platform is also available.

Google Forms Microsoft Forms

1 Consider Ptolemy's geocentric model. Determine whether the following statements are true.

(a) The background stars are fixed on the outermost sphere and move around the Sun. (T/F)

(b) Planets move in a small circle whose centre moves along a larger circle. (T/F)

2 Consider Copernicus' heliocentric model. Determine whether the following statements are true.

(a) The background stars are fixed on the outermost sphere and move around the Sun. (T/F)

(b) Mercury and Venus always appear close to the Sun because their orbits are inside the Earth's orbit. (T/F)

3 Consider both the Ptolemaic and Copernican models. Determine whether the following statements are true.

(a) Circular orbits of planets are assumed. (T/F)

(b) The Earth is at the centre of the Moon's orbit. (T/F)

4 Which of the following is an advantage of the Copernican model over the Ptolemaic model?

A The Copernican model can explain the retrograde motions of planets.

B The Copernican model can explain the changes in brightness of planets.

C The Copernican model is simpler.

D The Copernican model is more accurate.

Galileo Galilei 伽利略

Know more

Why was the geocentric model preferred?

Many people liked to think that the Earth was the centre of the universe. It gave humans a special status in the universe. Stars revolving around the Earth reinforced this belief and was supported by the Roman Catholic Church. Anyone who criticized the geocentric universe challenged the belief of the Church and risked attack. Besides, it was difficult for people to accept that Earth was travelling at a high speed in space and yet we remained on Earth. Because of these reasons, the Copernican model gained little support although most observations could be explained by the model.

Exercise 1.3

1 Which of the following statements about the retrograde motions of planets relative to the background stars is correct?

A The planets move from west to east throughout a night.

B The planets move from east to west over a period of time.

C The planets move at a changing rate.

D The paths of the planets must form a loop.

***2** Which of the following statements about the Copernican model is/are correct?

(1) It is less accurate than the Ptolemaic model in predicting positions of planets.

(2) It explains retrograde motions without using epicycles.

(3) The background stars lie on the outermost deferent, which rotates around the Earth.

A (1) and (2) only

B (1) and (3) only

C (2) and (3) only

D (1), (2) and (3)

3 Mars exhibits retrograde motion as shown in the figure below (Fig a). The orange line shows the path of Mars as observed from the Earth.

Fig a

(a) Draw an arrow in the figure to indicate the moving direction of Mars.

(b) Explain your answer in (a).

4 (a) In the Ptolemaic model, each planet moves uniformly on X. The centre of X moves uniformly around Y on a circle called Z. What are X, Y and Z?

(b) State the most revolutionary idea proposed by the Copernican model compared to the Ptolemaic model.

(c) State one common feature of the Ptolemaic and Copernican models.

***5** Venus is sometimes seen above the eastern horizon just before sunrise and is seen at other times above the western horizon just after sunset.

(a) Explain why we cannot see Venus in the sky at midnight.

(b) Suggest a reason why we cannot see Venus at noon.

***6** Figure b indicated the positions of the Earth and Saturn in their respective orbits at the same time instants.

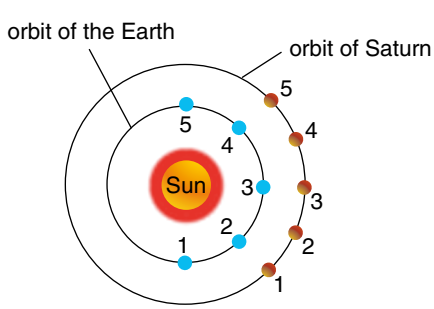

Fig b

(a) Determine whether Saturn exhibits prograde or retrograde motion when the Earth is passing (i) position 1 and (ii) position 3.

(b) How could the Copernican model explain the retrograde motion of Saturn?

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1.4 Galileo's astronomical discoveries

DSE goal
Describe Galileo's astronomical discoveries.

DSE exam
14(2)Q1.5

1 Galileo's observations

In 1609, more than sixty years after Copernicus's death, Galileo Galilei built his own telescope to study the sky (Fig 1.4a). He observed Venus through the telescope for three months and recorded the size and the shape of Venus he observed (Fig 1.4b).

Fig 1.4a *Galileo's telescope.*

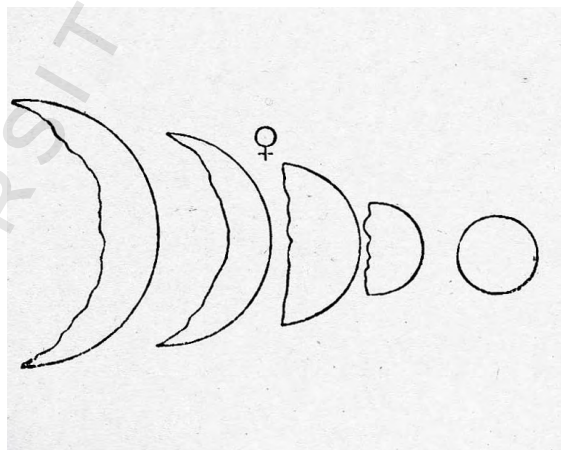


Fig 1.4b *Galileo's drawing of Venus he observed.*

✎ Science and technology are closely related. Very often a scientific discovery leads to some new technology. But in Galileo's work, we see how new technology led to some important scientific discoveries.

Galileo also observed other celestial objects. His astronomical discoveries supported the Copernican model. Some of his discoveries are listed below.

1 There are mountains and valleys on the Moon.

2 There are dark spots on the surface of the Sun.

3 There are four satellites orbiting Jupiter.

4 Venus undergoes a complete cycle of **phases** like the Moon.

Let us discuss the implications of these discoveries in the following sections.

phase 相位 / 相

Video

→ This video clearly shows the landscape of the Moon.

a The Moon

The Sun, the Moon and other celestial objects were once thought to be perfectly smooth spheres. However, by using his telescope, Galileo discovered that the Moon had mountains, valleys, and *craters* that are similar to those on the Earth (Fig 1.4c). It was not smooth at all.

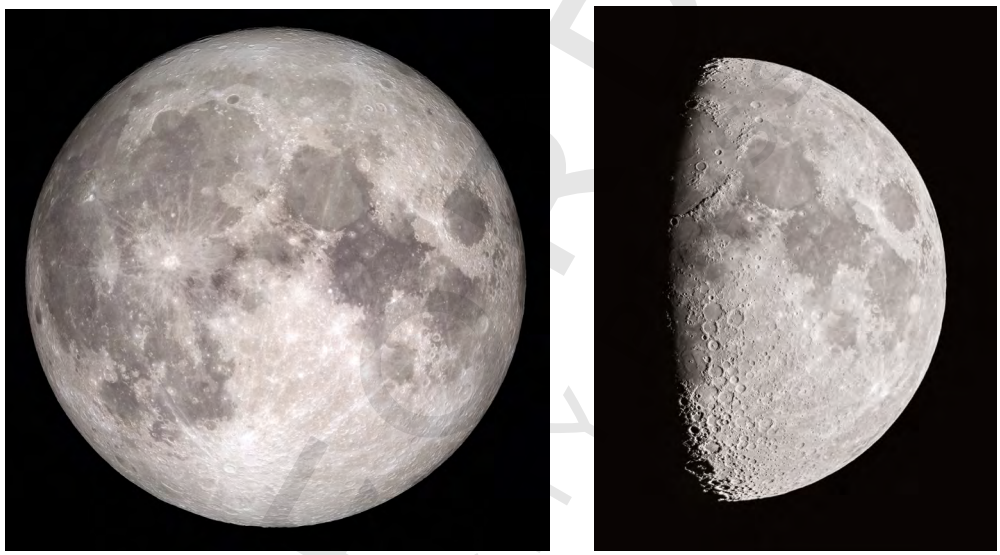


Fig 1.4c *The Moon as seen through a telescope. The Moon's landscape is particularly prominent when it is not full.*

Know more

The mysterious side of the Moon

Do you know that we can see only one side of the Moon from the Earth? Why is this so?

▸ p.44

✎ The dividing line of lunar day and night is called the terminator (晝夜線).

Never look at the Sun without suitable filters! You may seriously hurt your eyes, or even go blind.

b The Sun

Galileo also found imperfections on the surface of the Sun (Fig 1.4d). By using his telescope again, he discovered dark patches, which are known as **sunspots** today (Fig 1.4e).

By noting the changes in the positions of sunspots over time, Galileo inferred (correctly) that the Sun rotates, approximately once per month. This led Galileo to argue that the Earth may also rotate.

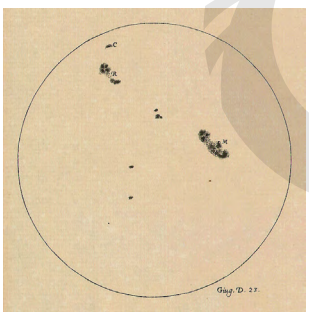
✎ A sunspot is a cool, dark area of the solar surface. The average spot is about twice the diameter of the Earth and may last for a week.

✎ The Sun is not a rigid body.

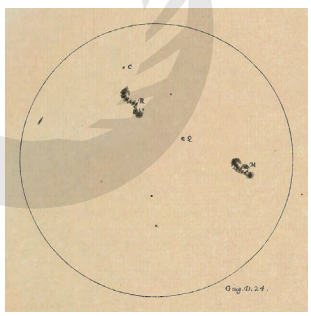
Experiment 1a

Sunspots and the rotation of the Sun (p.160)

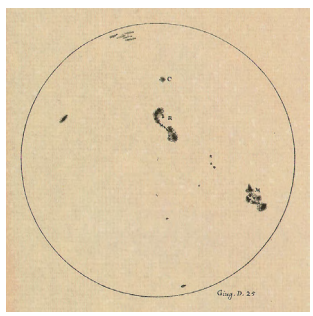
Useful link



(i) On 23 June 1613.



(ii) On 24 June 1613.



(iii) On 25 June 1613.

Fig 1.4d *Galileo's sketches on sunspots. The positions and shapes of the sunspots changed every day.*

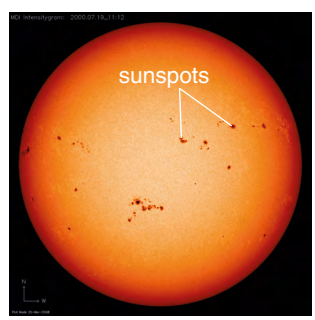


Fig 1.4e *Sunspots.*

Crater 隕石坑　　sunspot 太陽黑子

c Satellites of Jupiter

In 1610, Galileo also discovered four 'stars' moving around Jupiter (Fig 1.4f). These four 'stars' move relative to the background stars, but never move far from Jupiter. Each returns to the same position relative to Jupiter after a certain period of time.

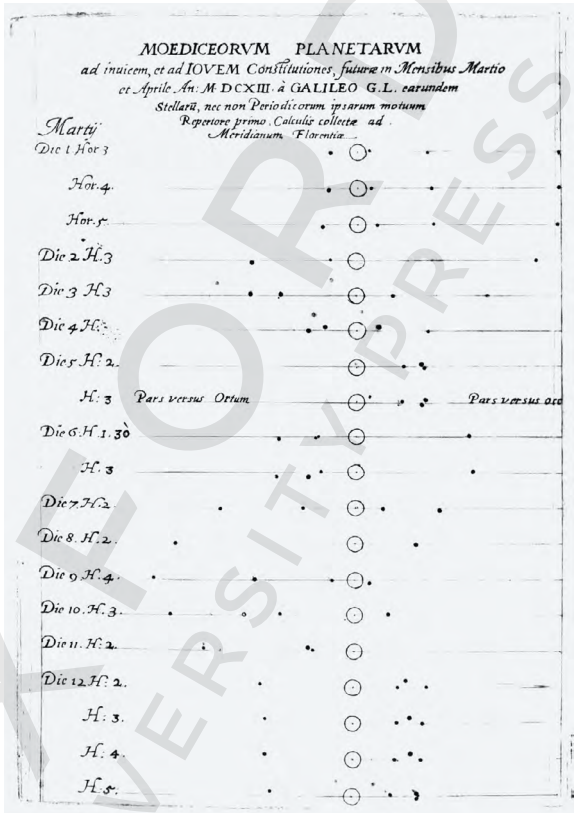


Fig 1.4f *Galileo recorded the positions of the stars (*) relative to Jupiter (○) on different days over the course of 12 days.*

Galileo realized that these 'stars' orbit around Jupiter. They are now known as the Galilean moons of Jupiter (Fig 1.4g). To Galileo, the fact that another planet had moons proved that the Earth was not the only centre of motion in the universe. This discovery provided strong evidence against the Ptolemaic model.

Fig 1.4g *Jupiter and its four Galilean moons taken with a modern telescope.*

The four Galilean moons are 1 Io, 2 Europa, 3 Ganymede, and 4 Callisto. The names are derived from the lovers of Zeus in Greek mythology.

✎ These four moons are the brightest moons of Jupiter. Actually, there are more than 90 moons of Jupiter.

d Phases of Venus

Galileo also found that Venus exhibited a complete cycle of phases, like those of the Moon (Fig 1.4h).

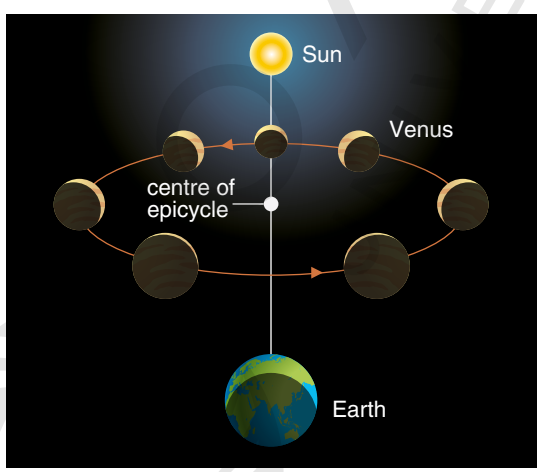
Fig 1.4h *Different phases of Venus over a period of time. The planet looks bigger when it is in its crescent phase.*

If the Ptolemaic model was correct, only the crescent phase of Venus could be observed.

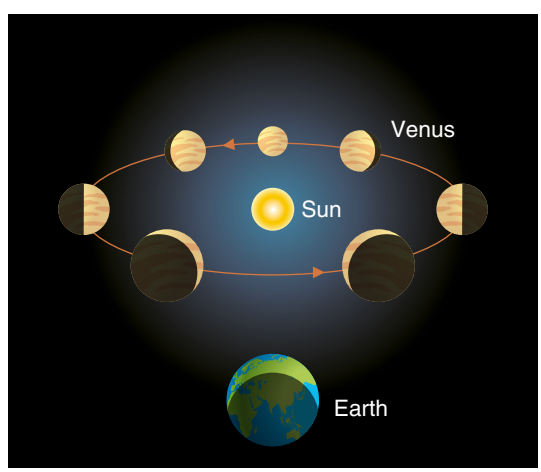
This finding directly contradicts the Ptolemaic model. In the Ptolemaic model, Venus does not go through the complete phase cycle. On the contrary, the heliocentric model can explain all the phases of Venus (Fig 1.4i). It also explains the connection between the phase and the apparent size of the planet. Venus is the furthest away from the Earth and appears the smallest when it shows a full phase.

You may watch a video showing the orbits of Venus and the Earth around the Sun, and the corresponding phases of Venus.

Useful link



(i) In the Ptolemaic model, Venue does not go through the complete phase cycle.



(ii) In the heliocentric model, Venus shows a complete cycle of phases.

Fig 1.4i *Phases of Venus in (i) the Ptolemaic model and (ii) the heliocentric model.*

The above observations implied that not all celestial bodies orbited the Earth. At least, Venus orbited the Sun. The discovery of the phases of Venus revealed a fundamental flaw of the Ptolemaic model.

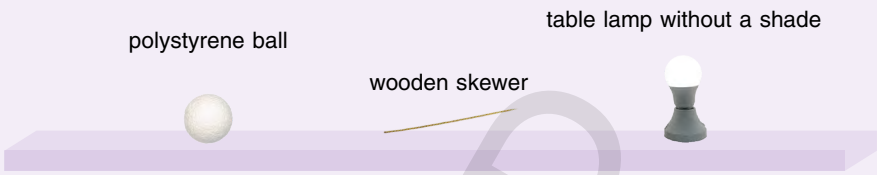
Video

→ This video demonstrates the Hands-on activity.

Hands-on activity

Different appearances of Venus

What you need



What to do

Use the sharp wooden skewer with care. Be careful of the hot lamp.

1

Stick a skewer into a polystyrene ball.

2

Orbit the ball around the lamp in a dark room.

What you find

Look at the ball. How do the illuminated parts of the ball look at different positions?

✎ Refer to Know more below.

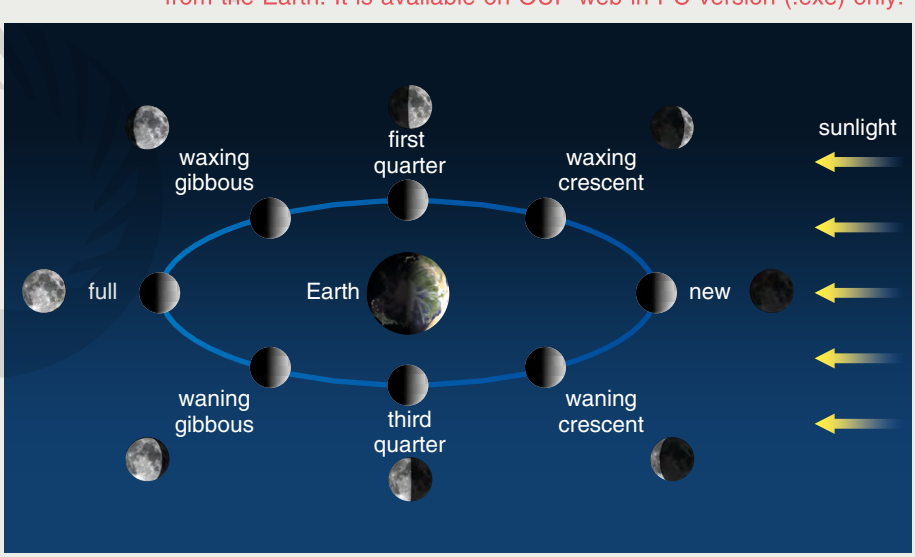
Know more

Phases of the Moon

The Moon's appearance undergoes a regular cycle of phases every 29.5 days. The figure below shows how the Moon phase changes with the Moon's position. Half of the Moon is always illuminated by sunlight. As the Moon moves in its orbit around the Earth, we see different portions of the lit side. These shapes are called phases.

✎ The website below provides an animation showing how the Moon's appearance changes throughout the month and a tool to see the current Moon phase.

Simulation → This simulation explains why the Moon shows different phases as seen from the Earth. It is available on OUP web in PC version (.exe) only.



Example 2 Position of Venus in its orbit

Several photographs of Venus were taken on different days over a period of time (Fig a).

Fig a

(a) Refer to Figure b. *X*, *Y* and *Z* are three positions in Venus' orbit. Determine the direction of movement of Venus around the Sun.

(b) What is the position of Venus, *X*, *Y* or *Z*, when its new phase is observed? Explain.

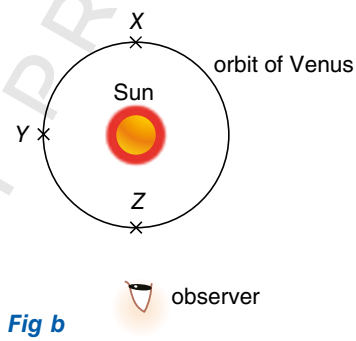


Fig b

Solution

(a) *X* ➔ *Y* ➔ *Z*

(b) *Z*.

The observer is on the opposite side of Venus as the Sun is.

Google Forms Microsoft Forms

Quick check 6

→ E-version in OUP Exercise Platform is also available.

1 Which of the following was **not** discovered on the Moon by Galileo?

A Valleys
B Seas
C Craters
D Mountains

2 Which of the following observations by Galileo showed that there was at least one planet orbiting the Sun?

A The mountains on the Moon
B The satellites of Jupiter
C The complete phase cycle of Venus
D The sunspots

Learn by practice 2

1 Galileo discovered the complete cycle of phases of Venus.

(a) Which model, the Ptolemaic or the Copernican, cannot explain this discovery?

(b) Suggest a phase of Venus that is not allowed by the model in (a)?

2 According to the heliocentric model, what is the position of Venus in Figure a when its full phase is observed from the Earth?

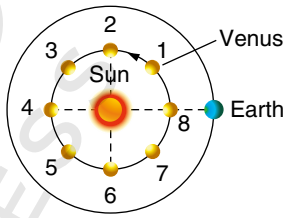


Fig a

DSE goal

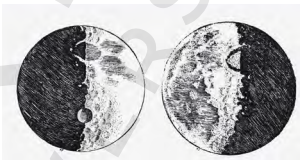
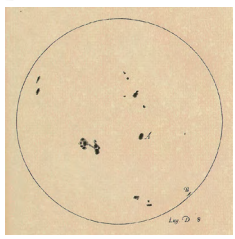
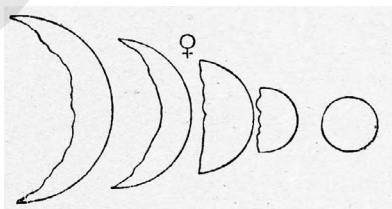
Discuss the implications of Galileo's astronomical discoveries.

DSE exam

15(2)Q1.4, 21(2)Q1.3

2 Implications of Galileo's discoveries

Table 1.4a summarizes Galileo's astronomical discoveries and their implications.

Discovery	Implication
There are mountains, valleys and craters on the Moon. 	• The Moon is not perfectly smooth. • The thought that celestial objects were perfectly smooth spheres is not correct.
On the surface of the Sun, there are sunspots that change their positions over time. 	• The Sun is not perfect. • The Sun rotates, so may the Earth.
There are four moons orbiting around Jupiter.	• The Earth is not the only centre of motion of the stars and planets. • It is possible that the Earth itself orbits something else.
Venus shows a complete cycle of phases. 	• This observation directly contradicts the Ptolemaic model. • It can only be explained by assuming that Venus orbits the Sun, thus providing strong evidence for the heliocentric model.

These moons follow Jupiter which is moving at high speed around the Sun. This implies that even as Earth moves at high speed around the sun, we can stay on Earth as well.

Table 1.4a *Galileo's astronomical discoveries and implications.*

Exercise 1.4

1 What of the following can be deduced from Galileo's astronomical observations?

A The Sun rotates.

B Jupiter has satellites.

C Venus orbits around the Sun.

D All of the above.

***2** Which diagram below shows the phase of Venus that could be seen from the Earth if the Ptolemaic model was correct?

(1) (2) (3)

A (2) only B (3) only

C (1) and (2) only D (2) and (3) only

***3** The figure shows the positions of the Earth, the Sun and Venus at a certain time (Fig a).

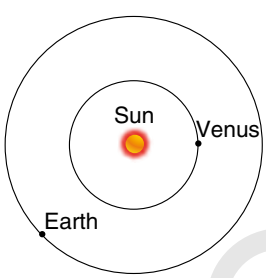


Fig a

Which diagram below best shows the Venus as seen from the Earth.

A B

C D

4 (a) Galileo observed sunspots with his telescope and made a conclusion about the motion of the Sun. What was it?

(b) State a precaution that should be taken when observing the Sun.

***5** How did the invention of the telescope support the development of astronomy? Give an example.

***6** (a) Why is the finding of Galileo's observation of the phases of Venus contradictory with the Ptolemaic model?

(b) Why does Venus look bigger in crescent phase?

***7** Galileo discovered four moons of Jupiter. How did this observation challenge the old belief?

Know more

Galileo Galilei (1564–1642)

Galileo did not invent a telescope, but he was the first to record the observations of celestial objects with a telescope. He based his arguments on his telescopic observations, rather than on philosophical motivations like the ancient Greeks did. Galileo's telescopic approach marked a new era of astronomy, and for this reason he is regarded as the father of observational astronomy.

OUP eResources

Galileo and the Roman Catholic Church

Galileo was one of the astronomers who supported the Copernican model. He published a book about his discoveries and his ideas of a heliocentric universe. Unfortunately, this contradicted the Church's geocentric belief. Galileo was accused of being a *heretic* at the age of 69 and was sentenced to house arrest for the rest of his life.

Galileo's trial for heresy by the Roman Catholic Church.

heretic 犯異端罪者

Key concepts 1

1 Size of celestial bodies

satellite < planet < star < nebula, star cluster < galaxy < galaxy cluster < supercluster of galaxies

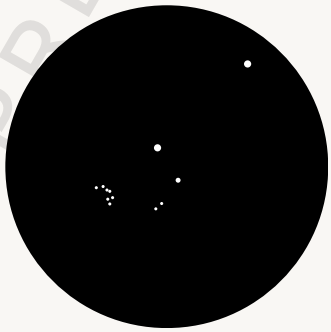
10^3 10^5 10^7 10^9 10^{11} 10^{13} 10^{15} 10^{17} 10^{19} 10^{21} 10^{23} 10^{25} size / m

Moon (satellite)

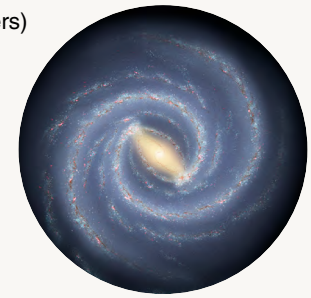
Sun (star)

Earth (planet)

Eagle Nebula (nebula)


galaxy cluster

M80 (star clusters)


Milky Way Galaxy (galaxy)

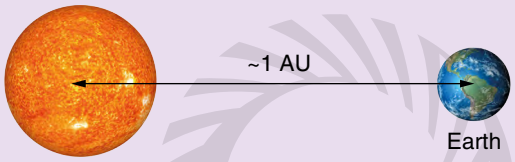
supercluster of galaxies

2 Unit of length

Astronomical unit (AU)

1 AU = 1.50×10^{11} m

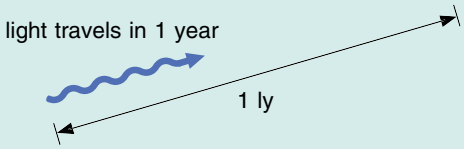
Average distance between the Earth and the Sun ≈ 1 AU



Light year (ly)

1 ly = distance travelled by light in a vacuum in one year

1 ly = 9.46×10^{15} m



Parsec (pc)

1 pc = 3.09×10^{16} m
= 3.26 ly

∴ 1 AU < 1 ly < 1 pc

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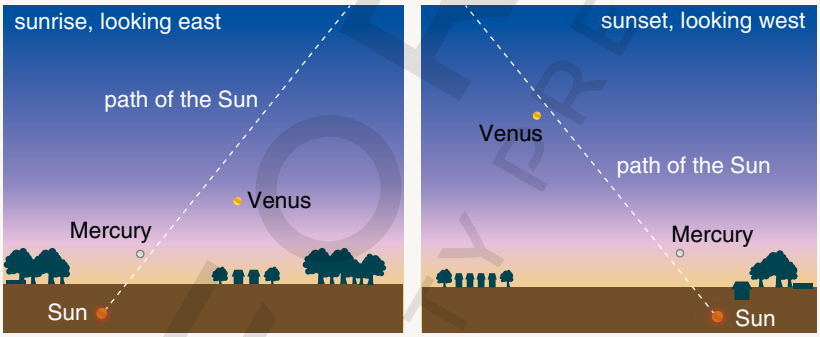
3 Development of our understanding of the universe

Ancient times

Early times observations

1 Retrograde motion

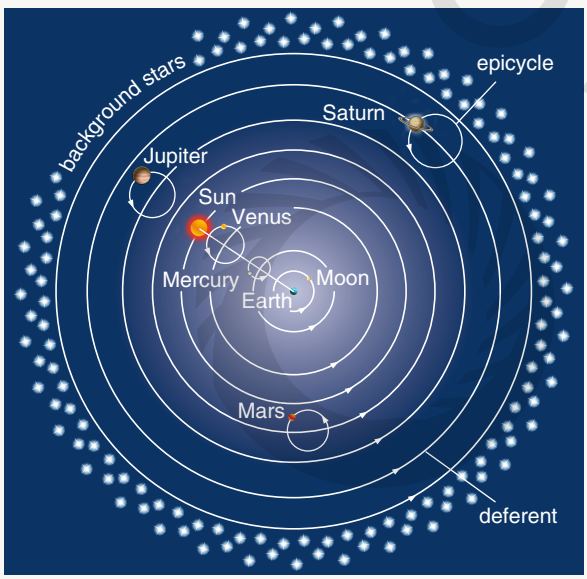
2 Motion of Mercury and Venus



168 AD

Ptolemaic geocentric model

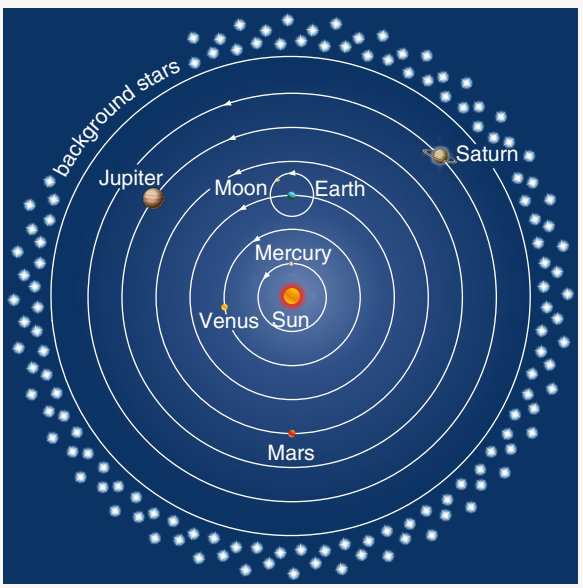
- Earth-centred
- Circular orbits — deferents & epicycles



1543

Copernican heliocentric model

- Sun-centred
- Circular orbits



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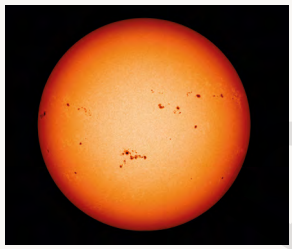
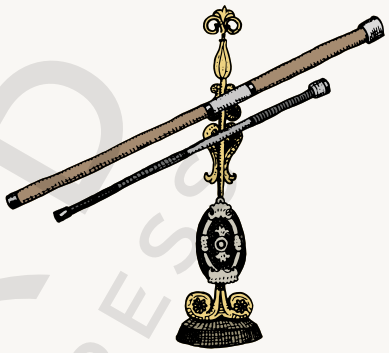
1609–1611

Galileo's discoveries

1 Mountains, valleys and craters on the Moon

2 Sunspots on the Sun





3 Jupiter's moons

4 Complete cycle of the phases of Venus

Key terms

1 astronomical unit (AU) 天文單位 p.11
2 celestial body 天體 p.2
3 deferent 均輪 p.19
4 epicycle 本輪 p.19
5 galaxy 星系 p.4
6 galaxy cluster 星系團 p.5
7 geocentric model 地心模型 p.19
8 heliocentric model 日心模型 p.21
9 light year (ly) 光年 p.12
10 nebula (pl. nebulae) 星雲 p.4
11 parsec (pc) 秒差距 p.13
12 phase 相位 / 相 p.27
13 planet 行星 p.3
14 prograde motion 順行 p.17
15 retrograde motion 逆行 p.17
16 satellite 衛星 p.3
17 star 恆星 p.2
18 star cluster 星團 p.4
19 sunspot 太陽黑子 p.28
20 supercluster 超星系團 p.5

Revision 1

Unless otherwise specified, take
the speed of light $c = 3.00 \times 10^8$ m s^{-1},
1 AU = 1.50×10^{11} m, 1 ly = 9.46×10^{15} m,
1 pc = 3.09×10^{16} m = 3.26 ly = 206 265 AU.

Multiple-choice questions

1 1.1 Which of the following statements about a nebula (Fig a) is correct?

Fig a

(1) It is the birthplace of stars.
(2) It is smaller than a galaxy.
(3) It has a solid structure.

A (1) only B (3) only
C (1) and (2) only D (2) and (3) only

2 1.1 Which of the following statements about a star cluster (Fig b) is correct?

Fig b

A A star cluster is a small galaxy.
B A star cluster is an assembly of galaxies.
C There are many star clusters in a nebula.
D A star cluster is formed by stars bound by gravity.

3 1.2 Which of the following is/are (a) unit(s) of time?

(1) Second
(2) Parsec
(3) Light year

A (1) only B (3) only
C (1) and (3) only D (2) and (3) only

4 1.3 Which of the following statements is/are correct?

(1) In both Ptolemaic and Copernican models, the full phase of Venus can be seen on Earth.
(2) The Earth is at the centre of the Moon's orbit in both Ptolemaic and Copernican models.
(3) Both Copernicus and Ptolemy thought that each planet moved in a circular orbit.

A (1) only B (3) only
C (1) and (2) only D (2) and (3) only

***5** 1.2 Neptune is a planet of the solar system (Fig c). Its orbit around the Sun can be regarded as circular with a radius of roughly 30.0 AU and a period of about 160 years. Estimate the orbital speed of Neptune.

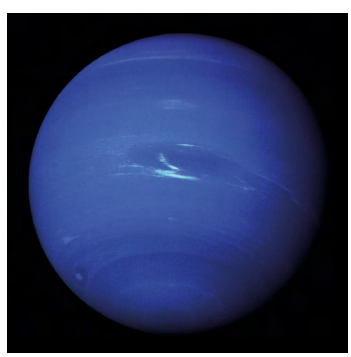


Fig c

A 0.188 AU yr^{-1} B 3.53×10^8 m s^{-1}
C 3210 km s^{-1} D 20 200 km h^{-1}

***6** 1.3 Which of the following statements about retrograde motion of planets is **incorrect**?

A Both the geocentric and heliocentric models can explain the retrograde motion of planets.
B Retrograde motion can only be explained by assuming the orbits of the planets are circular.
C A planet moves westwards in the sky when it undergoes retrograde motion.
D Saturn also exhibits retrograde motion as observed from the Earth.

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(For Q7–8.) Figure d shows parts of the solar system.

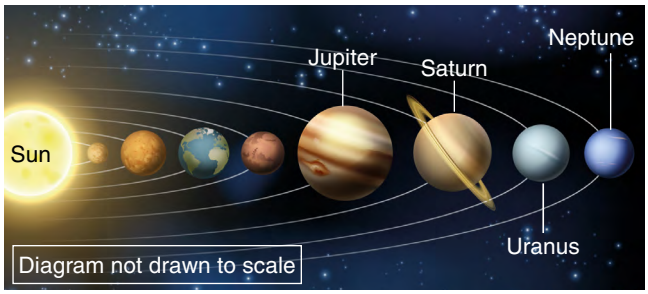

Fig d

****7** 1.3 As observed from Saturn, which planet(s) exhibit(s) retrograde motion?

(1) Jupiter
(2) Uranus
(3) Neptune

A (1) only
B (2) and (3) only
C (1), (2) and (3)
D None of them

Refer p.22

****8** 1.4 As observed from Saturn, which planet(s) show(s) a complete cycle of phases?

(1) Jupiter
(2) Uranus
(3) Neptune

A (1) only
B (2) and (3) only
C (1), (2) and (3)
D None of them

Refer p.30

9 1.2 **HKDSE 2016 Paper 2 Q1.6**

The explosion of a massive star towards the end of its life results in a supernova which appears extremely bright in the sky for some time. In 1987, a supernova (SN1987A) appeared and was visible to the naked eye in the Large Magellanic Cloud which is 163 000 light years away. In 1054, Chinese astronomers observed another supernova (SN1054) that appeared in the constellation of Taurus which is 6500 light years away. SN1987A happened about

A 933 years after SN1054.
B 155 567 years before SN1054. (53%)
C 156 500 years before SN1054.
D 162 067 years before SN1054.

10 1.3 **HKDSE 2015 Paper 2 Q1.5**

When the Earth lines up with the Sun and Mars as shown, how does Mars appear to move across the night sky as viewed from Earth?

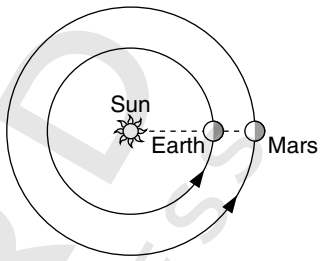

Fig e

A Mars moves from west to east against the background stars.
B Mars moves from east to west against the background stars. (40%)
C Mars does not move against the background stars.
D The movement of Mars cannot be determined because the east and west directions are not known.

11 1.3 **HKDSE 2018 Paper 2 Q1.3**

Which of the following observations can be explained by both the Ptolemy's geocentric model and the Copernicus's heliocentric model?

(1) The brightness of planets varies throughout the year.
(2) The retrograde motion of Mars in the sky.
(3) Venus shows a complete cycle of phases in the sky.

A (1) only
B (3) only
C (1) and (2) only (50%)
D (2) and (3) only

12 1.4 **HKDSE 2021 Paper 2 Q1.3**

Which of the following observations made by Galileo is/are consistent with the heliocentric model but not with the geocentric model of the universe?

(1) retrograde motion of Mars
(2) moons revolving around Jupiter
(3) changing phases of Venus

A (1) only
B (2) only
C (1) and (2) only
D (2) and (3) only (53%)

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Conventional questions

13 1.1 Figure f shows a celestial body called Titan which orbits around the planet Saturn.

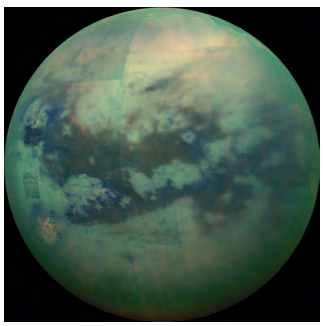

Fig f

(a) What kind of celestial body should Titan be classified as? (1 mark)

(b) Titan is a non-luminous object. How is it possible to be observed? (1 mark)

***14** 1.2 Perseverance is a Mars rover (Fig g). It took 202 days to travel 3.2 AU from the Earth to reach Mars. Find its average speed. Use the speed of light c to express your answer. (3 marks)

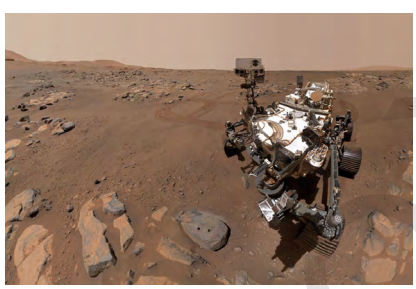

Fig g

***15** Misc The Milky Way Galaxy is a spiral galaxy which is about 100 000 to 120 000 light years in diameter.

(a) What is the range of diameter of the main disc of the Milky Way Galaxy in metres? (2 marks)

(b) The Sun is about 27 000 light years from the centre of the Milky Way Galaxy. Calculate the ratio of the orbital radius of the Sun to the average radius of the Milky Way Galaxy. (2 marks)

(c) Name one larger structure and one smaller structure relative to the Milky Way Galaxy. (2 marks)

(d) What unit(s) is/are commonly used to measure the distance/length in each of the structures in (c)? (2 marks)

***16** 1.3 The following figure shows the retrograde motion of Mars. The arrow shows the moving direction of Mars (Fig h).

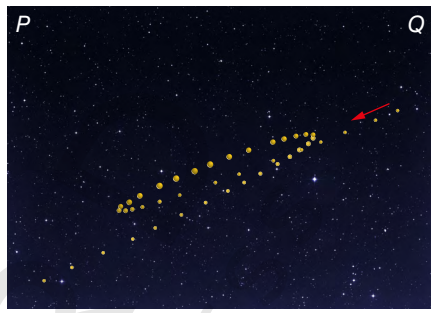

Fig h

(a) Explain the meaning of retrograde motion of Mars. (1 mark)

(b) Which side of the figure is the east, P or Q? (1 mark)

(c) Describe how the Ptolemaic model explains the retrograde motion of Mars. (2 marks)

(d) Describe, in terms of brightness and size, how Mars appears during retrograde motion when observed from the Earth. (2 marks)

***17** Misc Figure i shows a model of the universe.

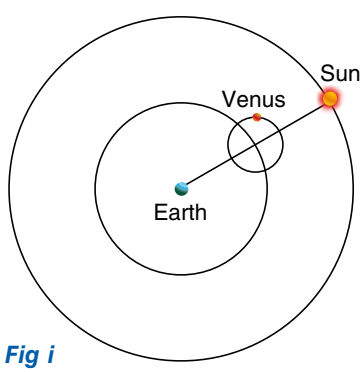

Fig i

(a) Is it a geocentric or heliocentric model? (1 mark)

(b) Draw a diagram to show the phase of Venus as seen from the Earth at the moment shown in Figure i. (1 mark)

(c) Explain how Galileo's finding on the phases of Venus conflicted with the above model. (2 marks)

(d) The heliocentric model can explain Galileo's finding on the phases of Venus as well as the connection between the phase and the apparent size of Venus. What is the connection? (2 marks)

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★18 1.3 Figure j shows a simplified Ptolemaic model.

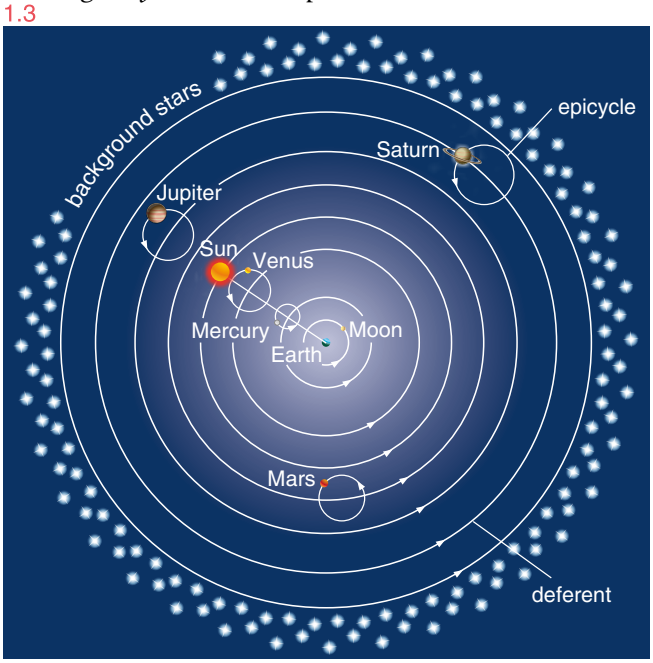

Fig j

(a) Explain the differences between epicycle and deferent in the Ptolemaic model. (2 marks)

(b) Where are the background stars located in this model? (1 mark)

(c) How does the motion of the background stars in this model differ from that in the Copernican model? (2 marks)

★★19 1.2 Fomalhaut is the brightest star in the constellation of Piscis Austrinus (Fig k). Light takes about 25 years to travel from the Earth to Fomalhaut.

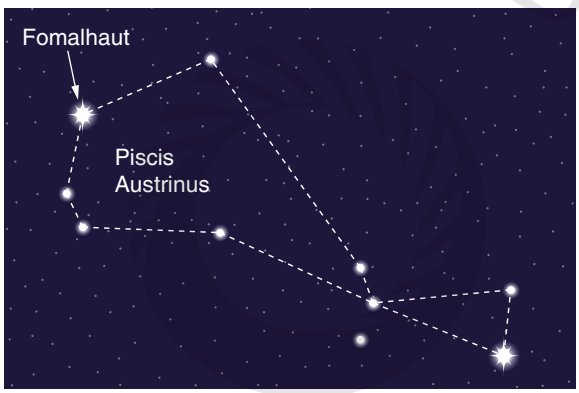

Fig k

(a) What is the meaning of one light year? (1 mark)

(b) What is the distance between the Earth and Fomalhaut in kilometres? (2 marks)

(c) A spacecraft can travel at an average speed of 20 km s^{-1}. Find the time required for the spacecraft to reach Fomalhaut from the Earth. Express the time in years. (Take 1 year = 3.16×10^7 s.) (3 marks)

(d) A planet revolves around Fomalhaut at a distance of roughly 120 AU with an orbital period of approximately 1700 Earth years. Find the planet's average orbital speed in km h^{-1}. (2 marks)

Refer p.12–13

★★20 Misc Figure l shows the orbits of Venus and the Earth. The numbers 1 to 5 on their orbits denote their positions at different times. For example, when Venus is at position 1 in its orbit, the Earth is also at position 1 in its orbit and so on.

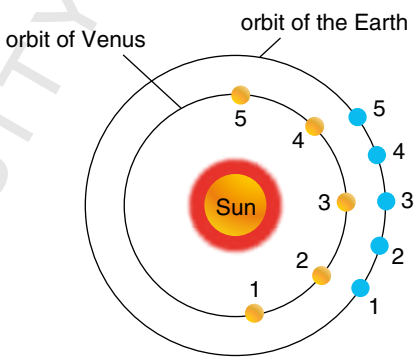

Fig l

(a) How does Venus move relative to the background stars most of the time? What is this motion called? (2 marks)

(b) For each of the positions shown above, draw the apparent position of Venus in the sky as observed from the Earth. (2 marks)

(c) When does Venus exhibit retrograde motion? (1 mark)

(d) Which of them, the Earth or Venus, has a longer period? Would it be possible to observe the retrograde motion of Venus if it had the same orbital period as the Earth? Why? (3 marks)

(e) At which position does Venus have the largest apparent size? (1 mark)

Refer p.22

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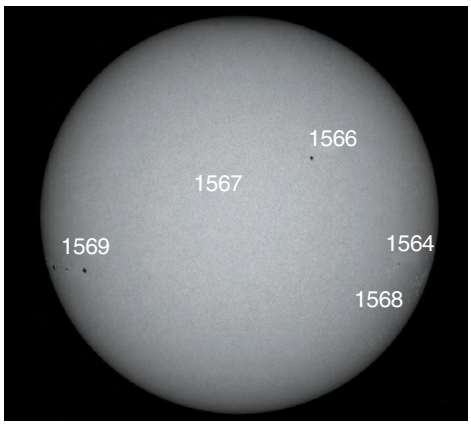
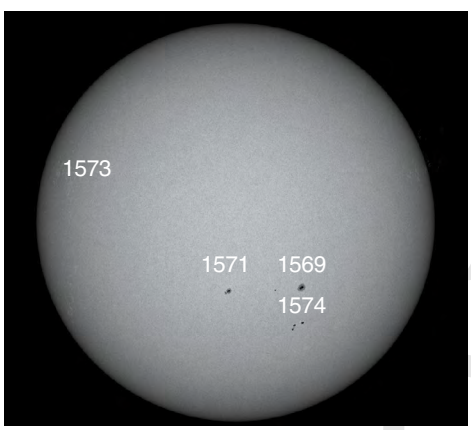
**21 1.4 Figure m shows two photographs of the Sun taken with a modern telescope. In these photographs, there are dark spots on the Sun's surface. Each spot is designated uniquely by the number next to them as shown.



(i) Taken on 11th September.



(ii) Taken on 17th September.

Fig m

(a) (i) Name the dark spots on the Sun. Who was the first person to discover these spots? (2 marks)

(ii) What were the implications of the discovery of these spots at that time? (2 marks)

(b) Estimate the rotation period of the Sun using the two photographs. [Hint: Analyze the motion of spot 1569] (2 marks)

(c) State the potential danger of observing these dark spots. Suggest a corresponding precaution. (2 marks)

Refer p.28

Nature of science

From observation to hypothesis (p.20)

Scientists often question the observations they make. They try to propose a possible answer to the question. This suggested answer is called a hypothesis. A hypothesis is built on the basis of observations and a few known facts. Further testing is needed to prove a hypothesis to be wrong. If the predictions made by a hypothesis contradict the observational data, scientists have to revise the hypothesis or try to develop an alternative. If the predictions made by a hypothesis agree with all observational data, it means that the hypothesis is likely to be true.

observation → question → hypothesis

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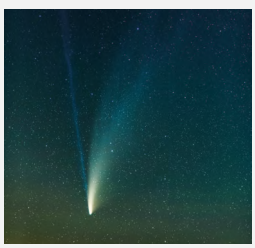
Know more

Small bodies in the solar system (p.3)

Small bodies are objects in the solar system that are neither the Sun, planets, dwarf planets, nor natural satellites. They also orbit the Sun. The following shows different kinds of small bodies in the solar system.

Comets

Comets are cosmic snowballs of frozen gases, rock, and dust that orbit the Sun. When a comet approaches the Sun, it is heated up. The ice in it vaporizes into gas. The gases and the dust form two tails that are typically millions of kilometers long.



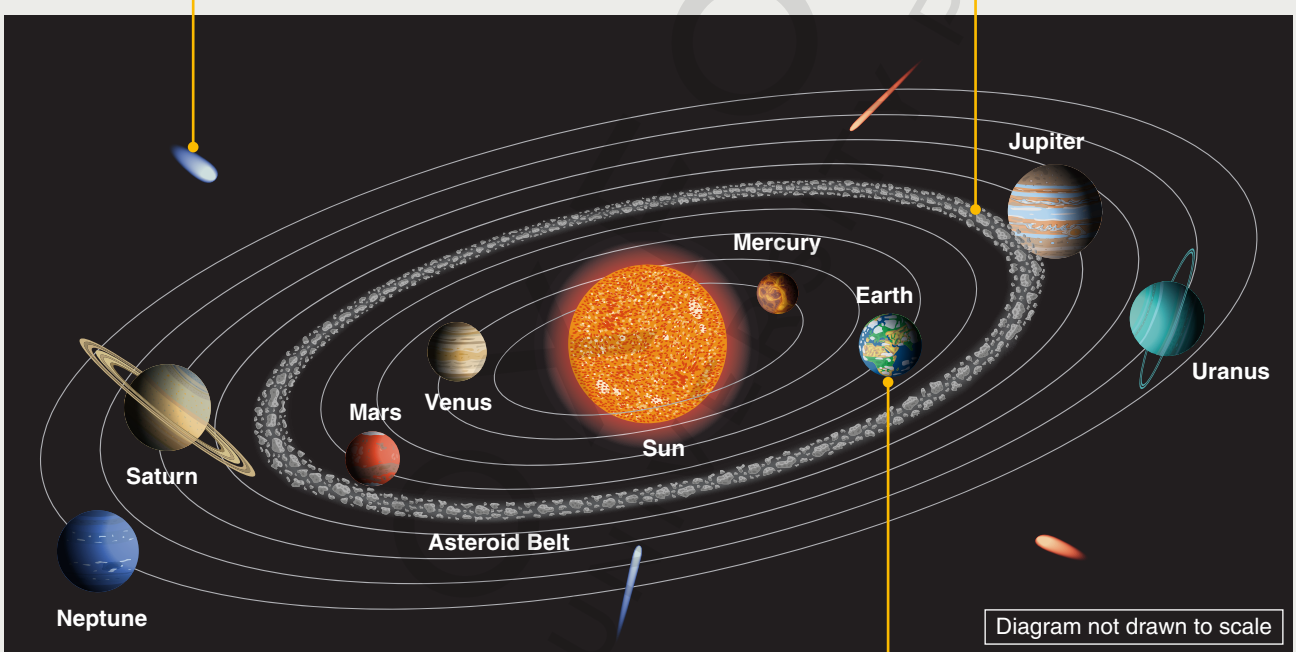
Comet NEOWISE.

Asteroids

Asteroids are rocky fragments with sizes ranging from about a few meters to hundreds of kilometers. Most of them are found in the *asteroid belt* between the orbits of Mars and Jupiter.

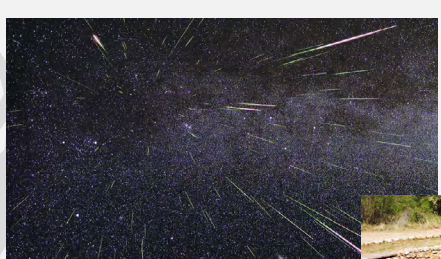
Asteroid 243 Ida.

✎ Asteroid 243 Ida is located in the asteroid belt. It has an average diameter of 31.4 km.



Meteoroids

Meteoroids are small rocky or metallic bodies in outer space. They are significantly smaller than asteroids. When they enter the Earth's atmosphere, they burn up and appear as bright paths called *meteors*. Any pieces that do not burn up completely and reach the ground are called *meteorites*.



Time lapse image of Perseid meteor shower.

Hoba, the largest meteorite ever found, in Namibia.

✎ The Hoba meteorite measures about 2.7 m across and weighs about 60 tons.

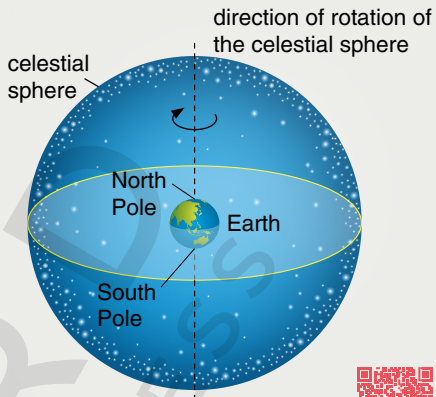
comet 彗星　asteroid 小行星　asteroid belt 小行星帶　meteoroid 流星體　meteor 流星　meteorite 隕石

Know more

Celestial sphere (p.16)

Ancient astronomers thought that the sky was a great sphere surrounding the Earth, and stars were firmly attached to the inside surface of the sphere. Such a sphere is called the celestial sphere. Stars move as the celestial sphere was rotating westwards around the fixed and unmoving Earth at the centre.

Today we know that it is the Earth instead of the sky that rotates. Furthermore, stars are not at the same distance from the Earth. However, the celestial sphere remains a useful model for understanding the movement of stars as seen from the Earth.



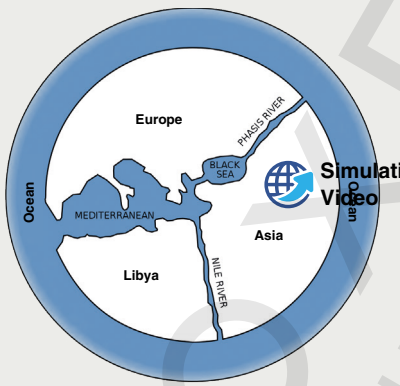
Simulation Video → This simulation explains the structure of the celestial sphere and the functions of different parts of the sphere model. It is available on OUP web in PC version (.exe) only.

→ This video is an introduction to the celestial sphere. It also explains the use of the sphere.

Astronomy of the ancient Greeks (p.19)

In ancient times, people in different civilizations were curious about the universe when they looked up at the sky. They wanted to find out whether the movement of the celestial bodies in the sky are regular and predictable. Among all the civilizations, the ancient Greeks used to discuss rationally. They tried to build a model that could explain the movement of the celestial bodies in the sky.

At the beginning, the ancient Greeks thought that the Earth was flat and surrounded by an ocean. Everyone at different locations on the Earth would see the same stars overhead.



The Earth was thought to be flat and surrounded by an ocean.

The universe was thought to be geocentric.

Later, astronomers realized that the Earth was indeed a sphere. It was suggested that the Earth was located at the centre of the universe and all other celestial bodies moved around the Earth uniformly in circular orbits.

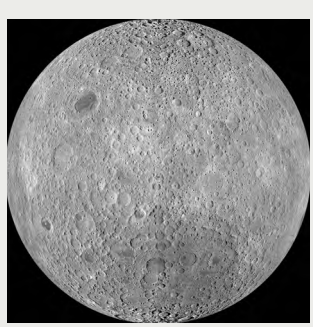
The mysterious side of the Moon (p.28)

You may be surprised to know that the Moon always faces the Earth with the same side. This side is called the near side of the Moon.

The reason for this is that the Moon rotates on its axis at the same rate that the Moon orbits the Earth. Once the Moon completes one full rotation on its axis, it also completes one orbit around the Earth.

The side facing away from Earth is called the far side of the Moon. It can only be seen by the human eye from a spacecraft. In 2019, the Chinese lunar rover *Chang'e 4* achieved the first ever soft-landing on the far side of the Moon in history.

Near side of the Moon.



Far side of the Moon.

Useful link